HARDWARE DESIGN PROJECT

CSS Physical Layer Transmitter

Design, RTL, Verification, FPGA & ASIC Implementation of a Chirp Spread Spectrum PHY Layer Transmitter

PROJECT TEAM

Mohab Elsayed

Mohammed Essam

Ali Elbadry

Khaled Gamal

Yahia Ayman

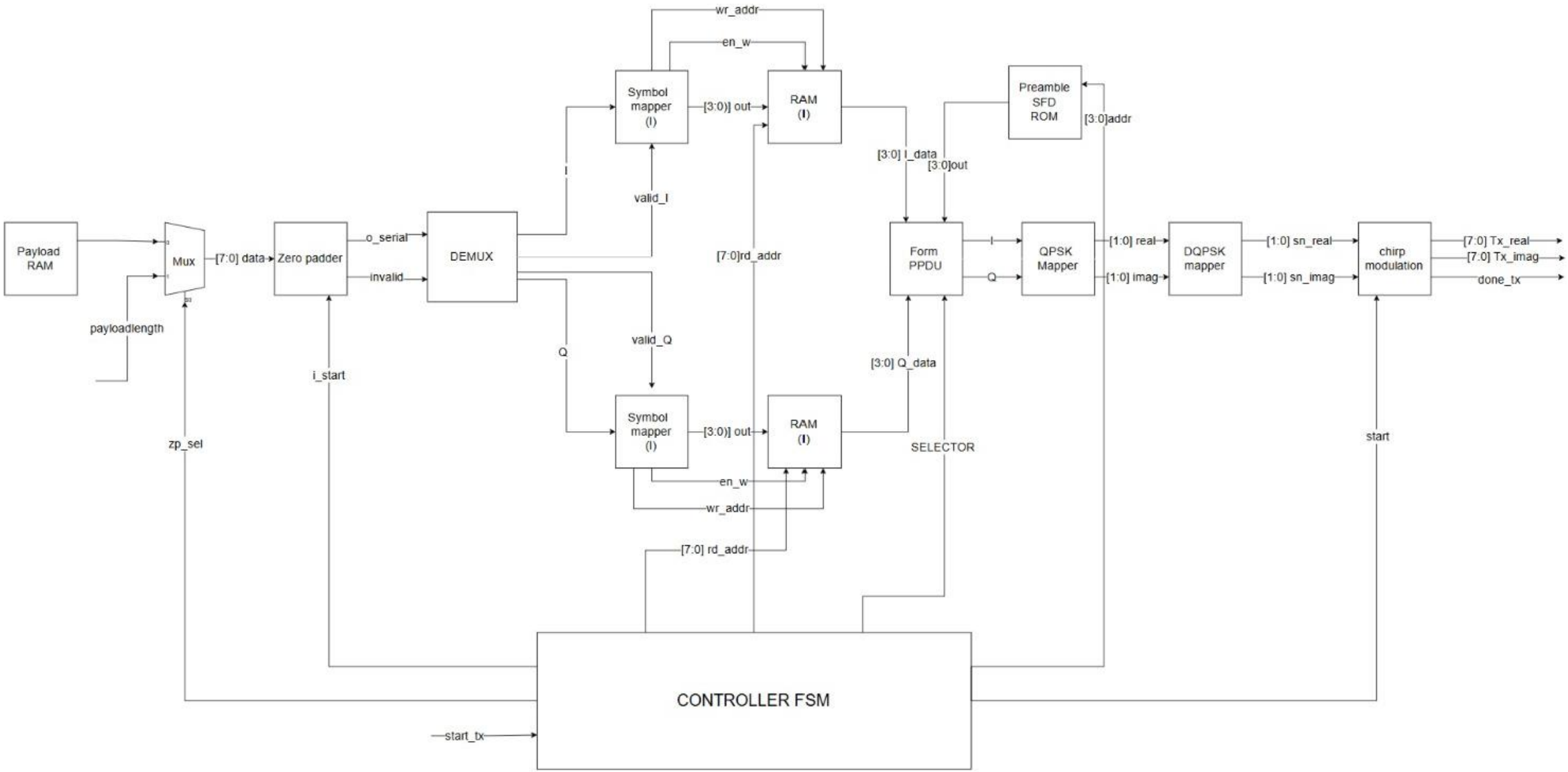
Muslim Elautey

Slides Overview

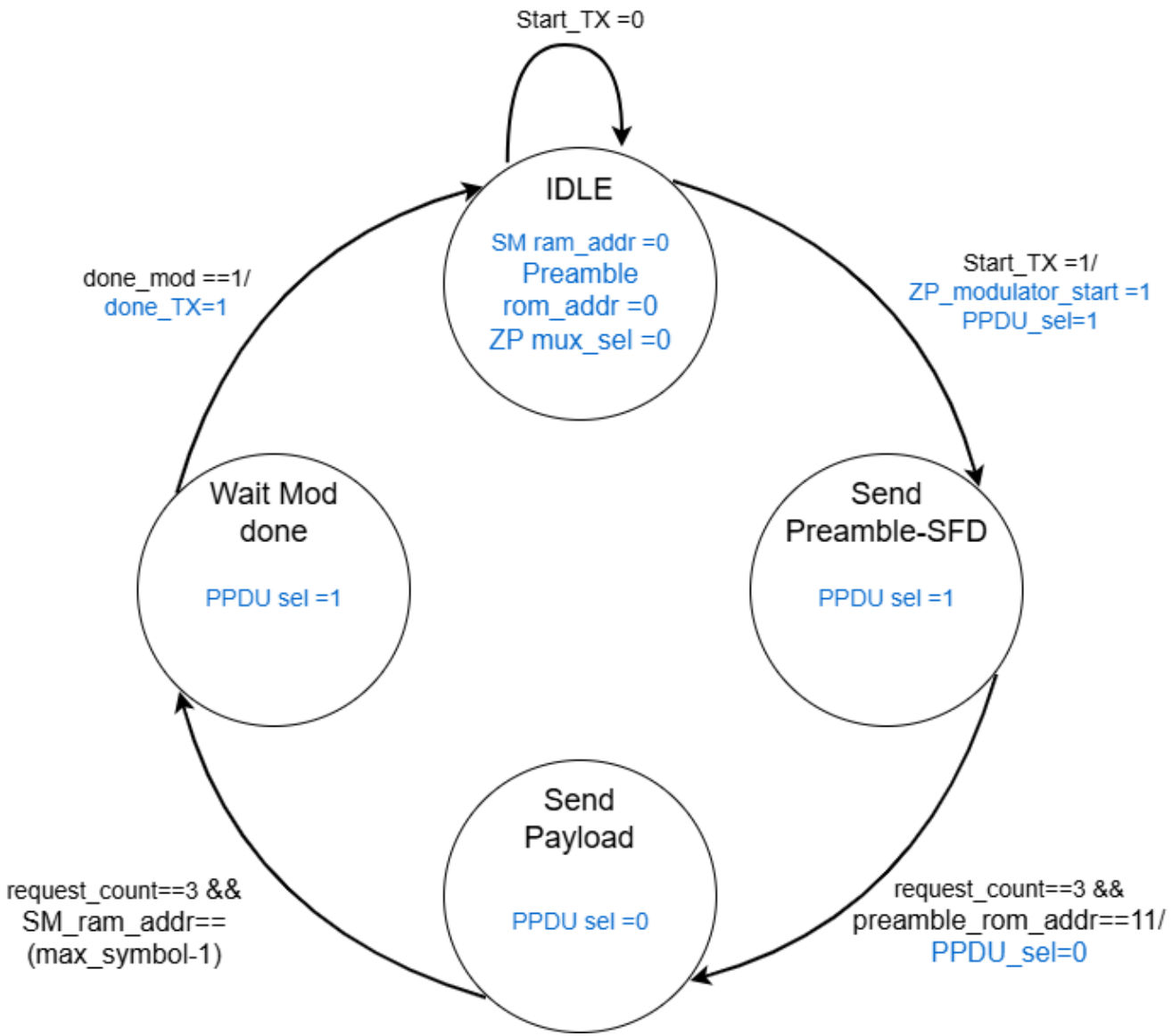
- Design
  -top level Architecture
  -FSM Diagram
  -Blocks Architecture
- Verification
  -MATLAB Model results
- FPGA
  -timing report results
  -resource utilization
- ASIC
  -Synthesis
  -DFT
  -Formality
  -PnR (ndm&design_library – floor plan – power plan – placement – CTS – Routing)

Design

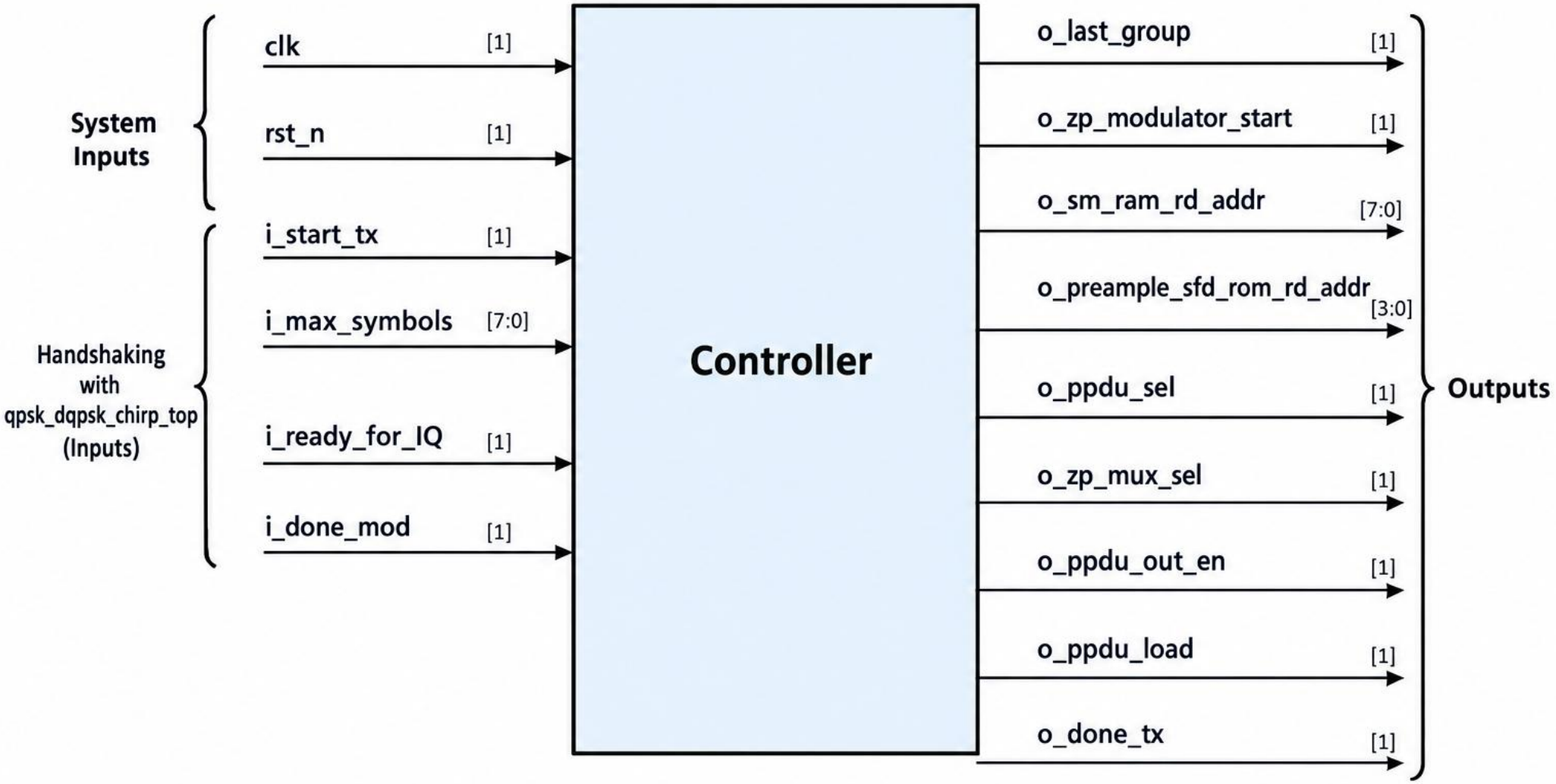
Top-Level Architecture



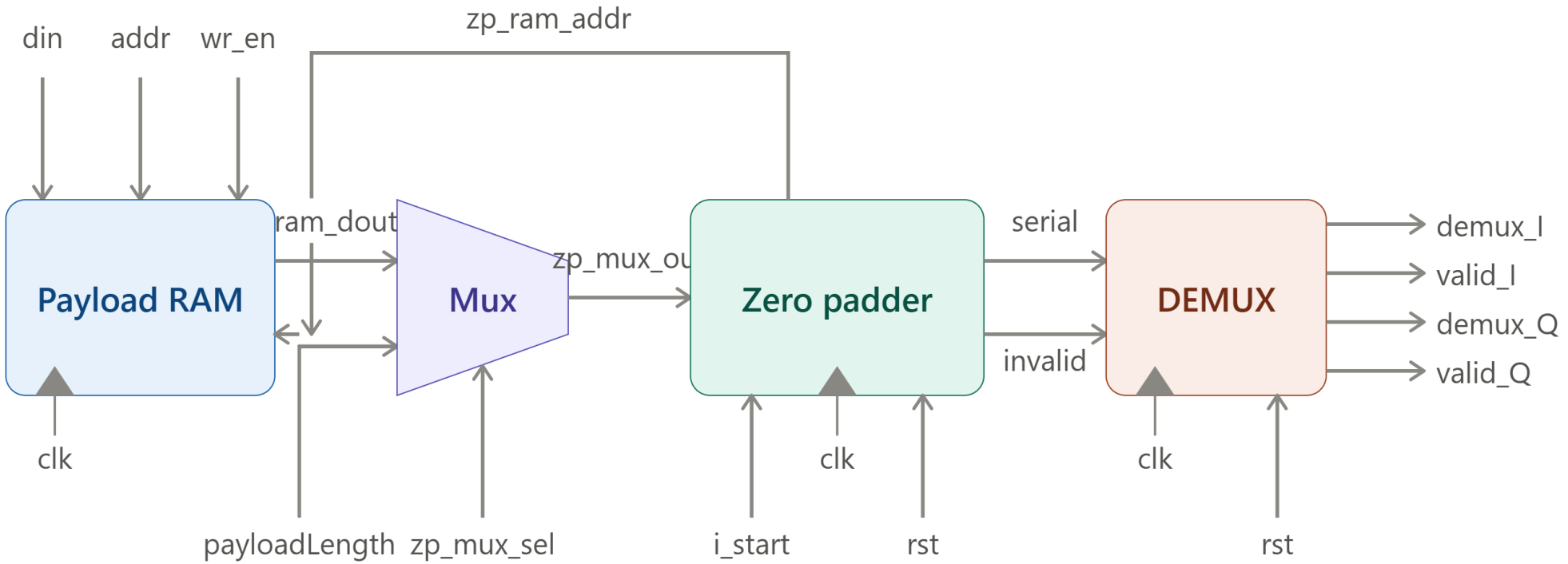
FSM Diagram


Start_TX =0
IDLE
SM ram_addr =0
Preamble
rom_addr =0
ZP mux_sel =0
Start_TX =1/
ZP_modulator_start =1
PPDU_sel=1
Send
Preamble-SFD
PPDU sel =1
request_count==3 &&
preamble_rom_addr==11/
PPDU_sel=0
Send
Payload
PPDU sel =0
request_count==3 &&
SM_ram_addr==
(max_symbol-1)
Wait Mod
done
PPDU sel =1
done_mod ==1/
done_TX=1

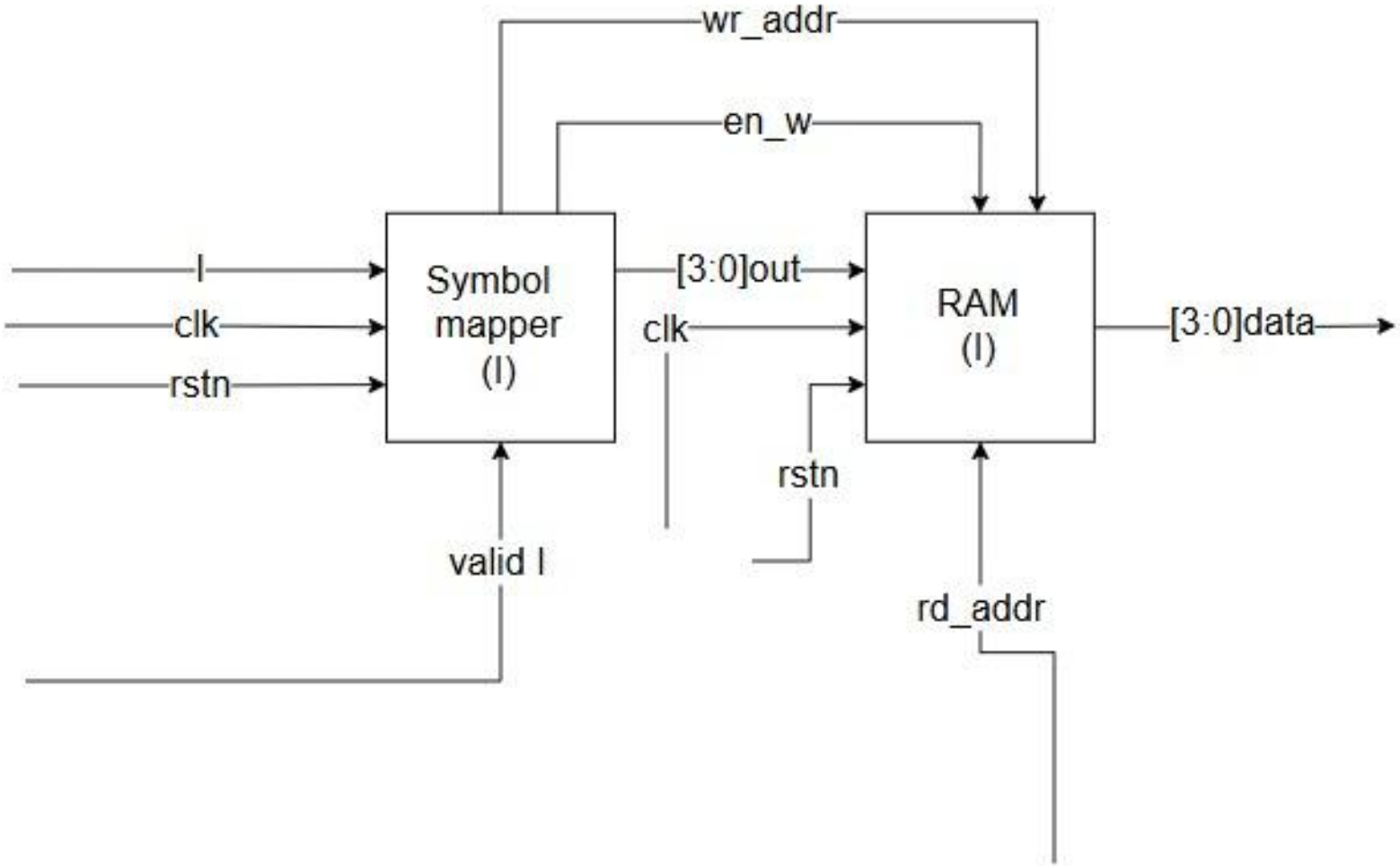
Stage 1


System Inputs
clk
[1]
rst_n
[1]
Handshaking with qpsk_dqpsk_chirp_top (Inputs)
i_start_tx
[1]
i_max_symbols
[7:0]
i_ready_for_IQ
[1]
i_done_mod
[1]
Controller
o_last_group
[1]
o_zp_modulator_start
[1]
o_sm_ram_rd_addr
[7:0]
o_preample_sfd_rom_rd_addr
[3:0]
o_ppdu_sel
[1]
o_zp_mux_sel
[1]
o_ppdu_out_en
[1]
o_ppdu_load
[1]
o_done_tx
[1]
Outputs

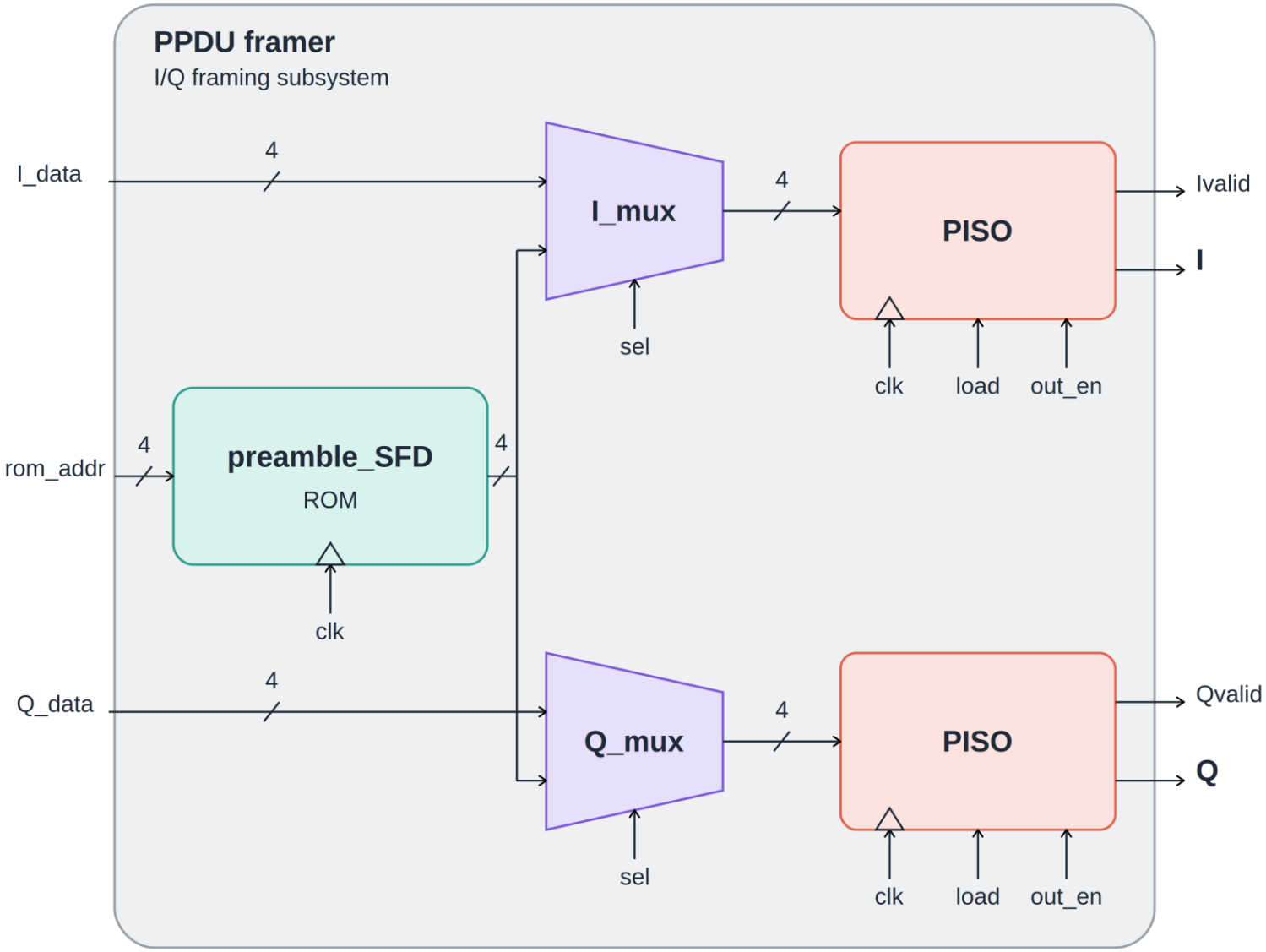
Stage 2


din
addr
wr_en
zp_ram_addr
Payload RAM
ram_dout
Mux
zp_mux_ou
Zero padder
serial
invalid
DEMUX
demux_I
valid_I
demux_Q
valid_Q
clk
clk
clk
payloadLength
zp_mux_sel
i_start
rst
rst

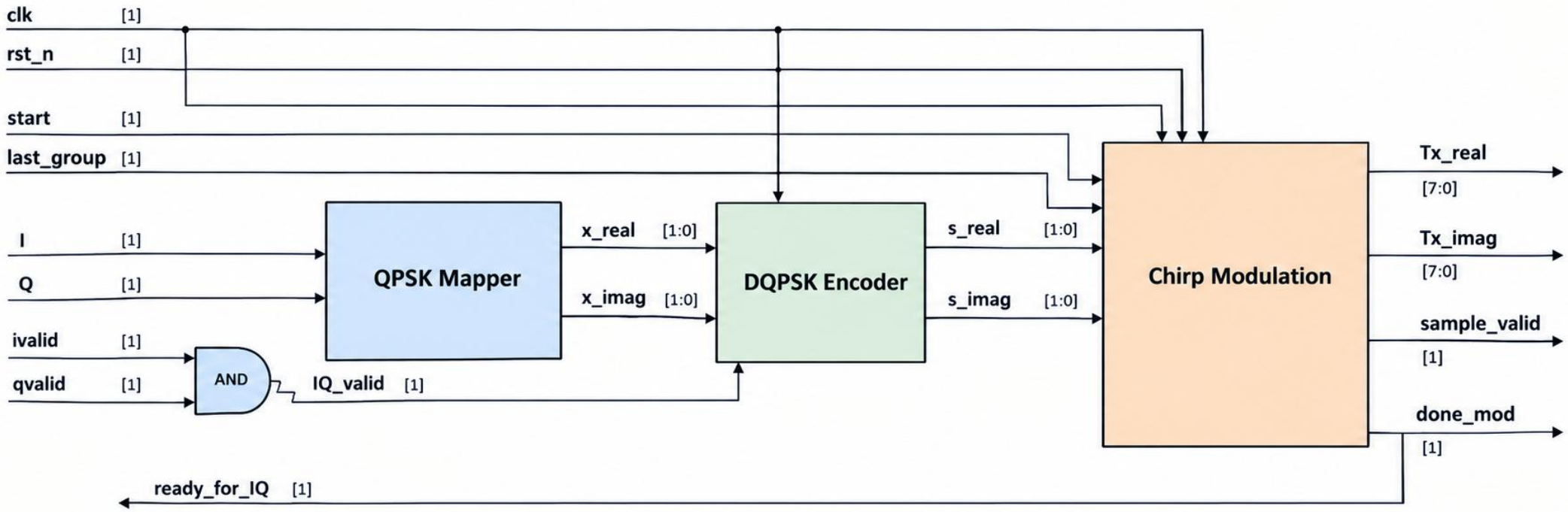
Stage 3


wr_addr
en_w
I
clk
rstn
Symbol
mapper
(I)
[3:0]out
clk
RAM
(I)
[3:0]data
rstn
valid I
rd_addr

Stage 4


PPDU framer
I/Q framing subsystem
I_data
4
I_mux
4
PISO
Ivalid
I
sel
clk
load
out_en
rom_addr
4
preamble_SFD
ROM
4
clk
Q_data
4
Q_mux
4
PISO
Qvalid
Q
sel
clk
load
out_en

Stage 5


clk [1]
rst_n [1]
start [1]
last_group [1]
I [1]
Q [1]
ivalid [1]
qvalid [1]
AND
IQ_valid [1]
QPSK Mapper
x_real [1:0]
x_imag [1:0]
DQPSK Encoder
s_real [1:0]
s_imag [1:0]
Chirp Modulation
Tx_real [7:0]
Tx_imag [7:0]
sample_valid [1]
done_mod [1]
ready_for_IQ [1]

Verification

Test_case from test_bench

```
// ================================================================== //
// TEST CASE 2: 25 Bytes Payload
// ================================================================== //
$display("\n--- Starting Test Case 2: 25 Bytes ---");
apply_reset();
payloadLength = 25;

// Read the 25-byte payload file
$readmemb("C:/iti/Zigbee/project_1/project_1.srcs/sources_1/new/mac_payload_raw_25B.txt", mac_payload_mem);

for (i = 0; i < 25; i = i + 1) begin
    write_payload_byte(i[7:0], mac_payload_mem[i]);
end

file_real = $fopen("C:/iti/Zigbee/project_1/project_1.srcs/sources_1/new/rtl_tx_real_25B.txt", "w");
file_imag = $fopen("C:/iti/Zigbee/project_1/project_1.srcs/sources_1/new/rtl_tx_imag_25B.txt", "w");

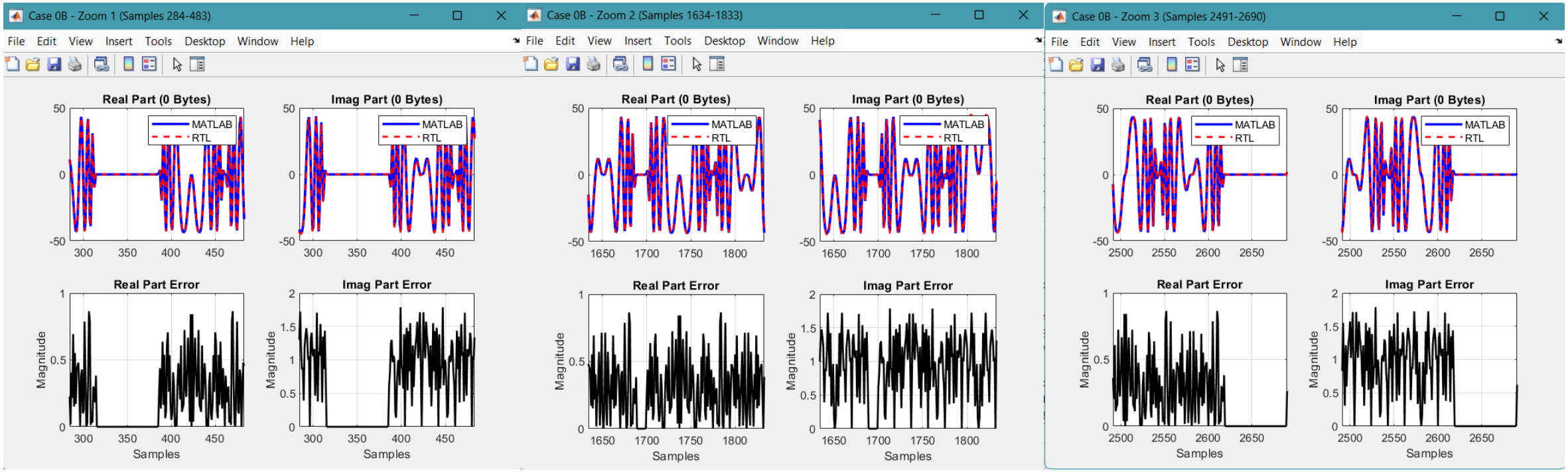
sampling_en = 1;
trigger_tx();
wait(done_Tx == 1'b1);
repeat(5) @(negedge clk);
sampling_en = 0;

$fclose(file_real);
$fclose(file_imag);
$display("Test Case 2 Completed!");

repeat(10) @(negedge clk);
```

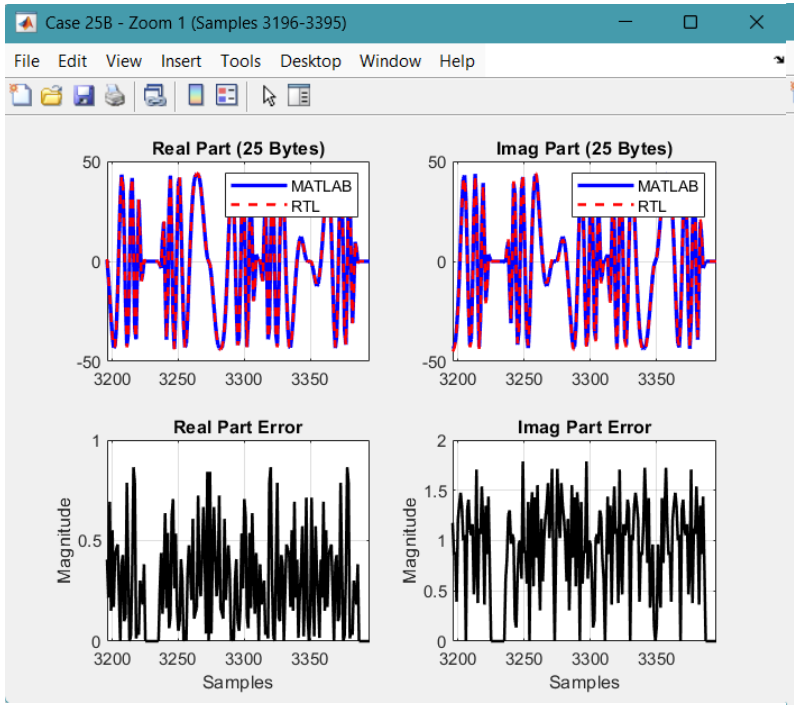
Results

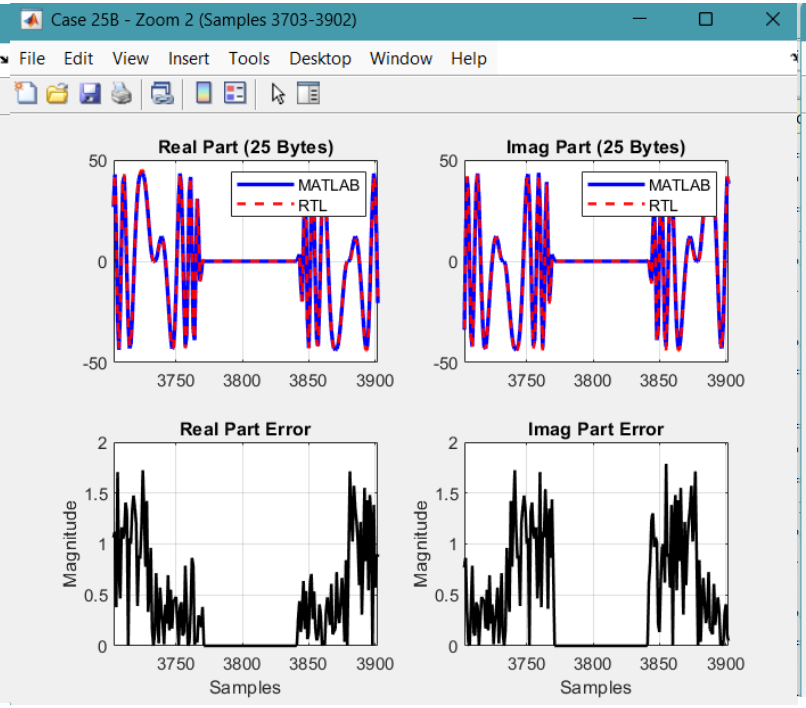
Case_1 (0 Bytes)

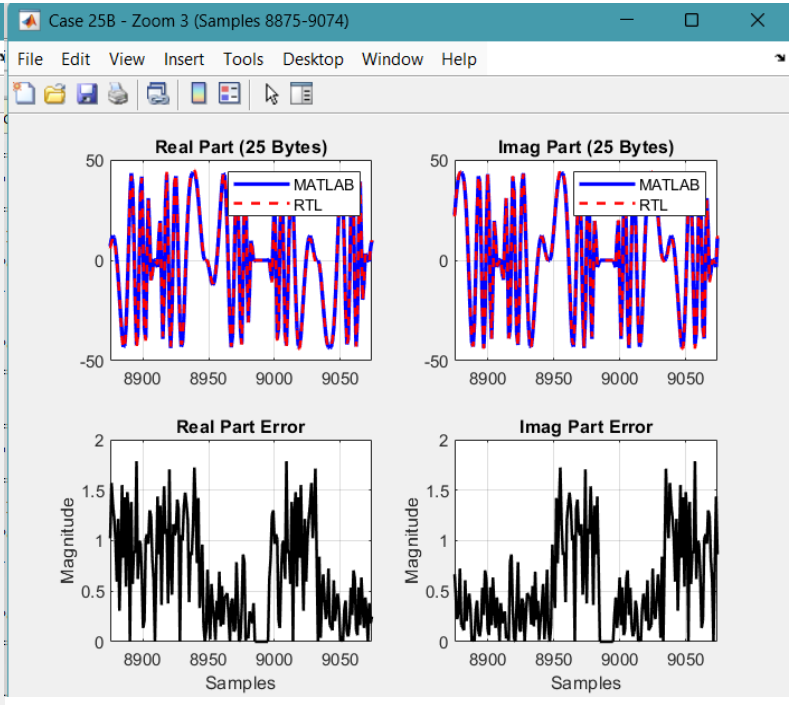


Results

Case_2 (25 Bytes)

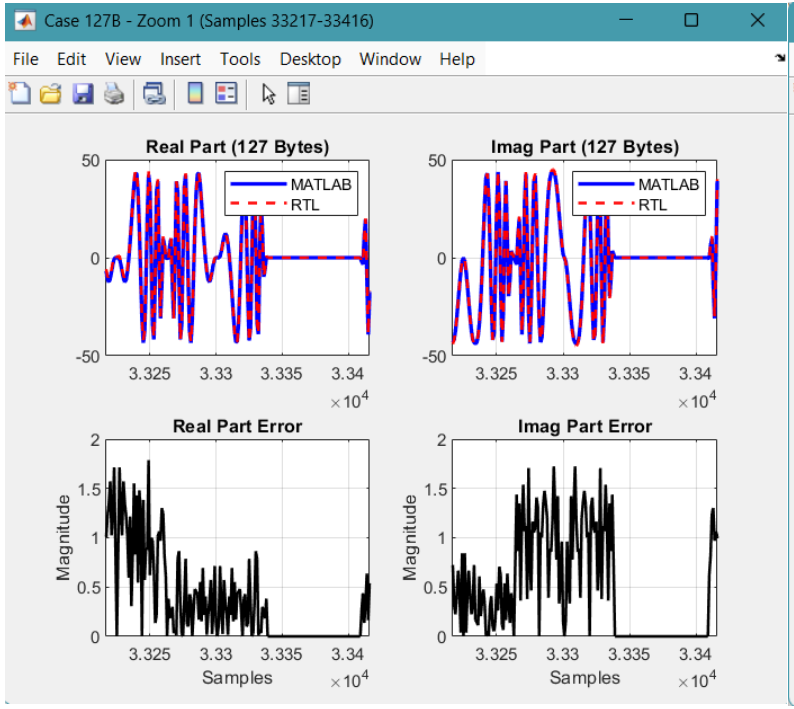


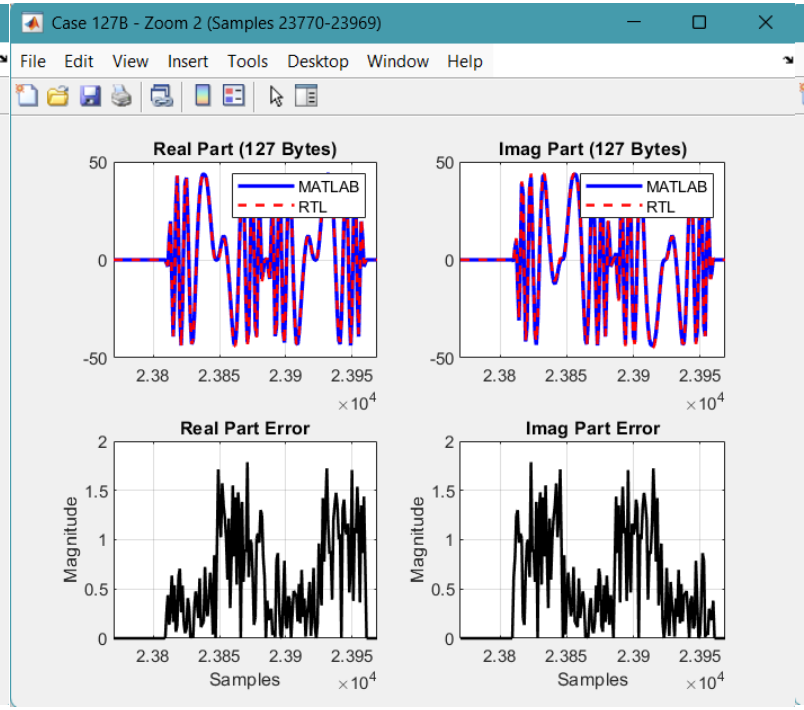


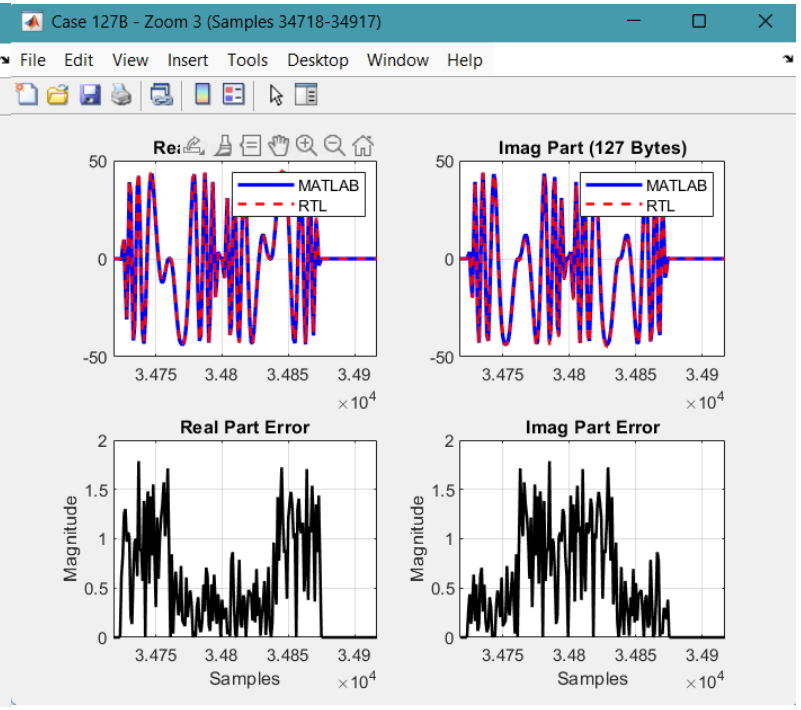


Results

Case_3 (127 Bytes)







Reports

```
==========================================
PROCESSING TEST CASE: 0 BYTES
==========================================
Average Error (Real)           = 0.24771
Average Error Percentage (Real)= 0.56508 %
------------------------------------------
Average Error (Imag)           = 0.77081
Average Error Percentage (Imag)= 1.7584 %
==========================================

==========================================
PROCESSING TEST CASE: 25 BYTES
==========================================
Average Error (Real)           = 0.42859
Average Error Percentage (Real)= 0.97769 %
------------------------------------------
Average Error (Imag)           = 0.57932
Average Error Percentage (Imag)= 1.3215 %
==========================================

==========================================
PROCESSING TEST CASE: 127 BYTES
==========================================
Average Error (Real)           = 0.49513
Average Error Percentage (Real)= 1.1295 %
------------------------------------------
Average Error (Imag)           = 0.51278
Average Error Percentage (Imag)= 1.1698 %
```

```
==========================================
OFFICIAL MSE VERIFICATION REPORT: 0 BYTES
==========================================
MSE (Real)           = 0.000212
MSE (Imaginary)      = 0.001571
MSE (Complex Total)  = 0.000891
------------------------------------------
Target Threshold     = 0.005
STATUS: PASSED (MSE is below the threshold!)
==========================================

==========================================
OFFICIAL MSE VERIFICATION REPORT: 25 BYTES
==========================================
MSE (Real)           = 0.000691
MSE (Imaginary)      = 0.001093
MSE (Complex Total)  = 0.000891
------------------------------------------
Target Threshold     = 0.005
STATUS: PASSED (MSE is below the threshold!)
==========================================

==========================================
OFFICIAL MSE VERIFICATION REPORT: 127 BYTES
==========================================
MSE (Real)           = 0.000877
MSE (Imaginary)      = 0.000905
MSE (Complex Total)  = 0.000891
------------------------------------------
Target Threshold     = 0.005
STATUS: PASSED (MSE is below the threshold!)
==========================================
```

Lint

Report

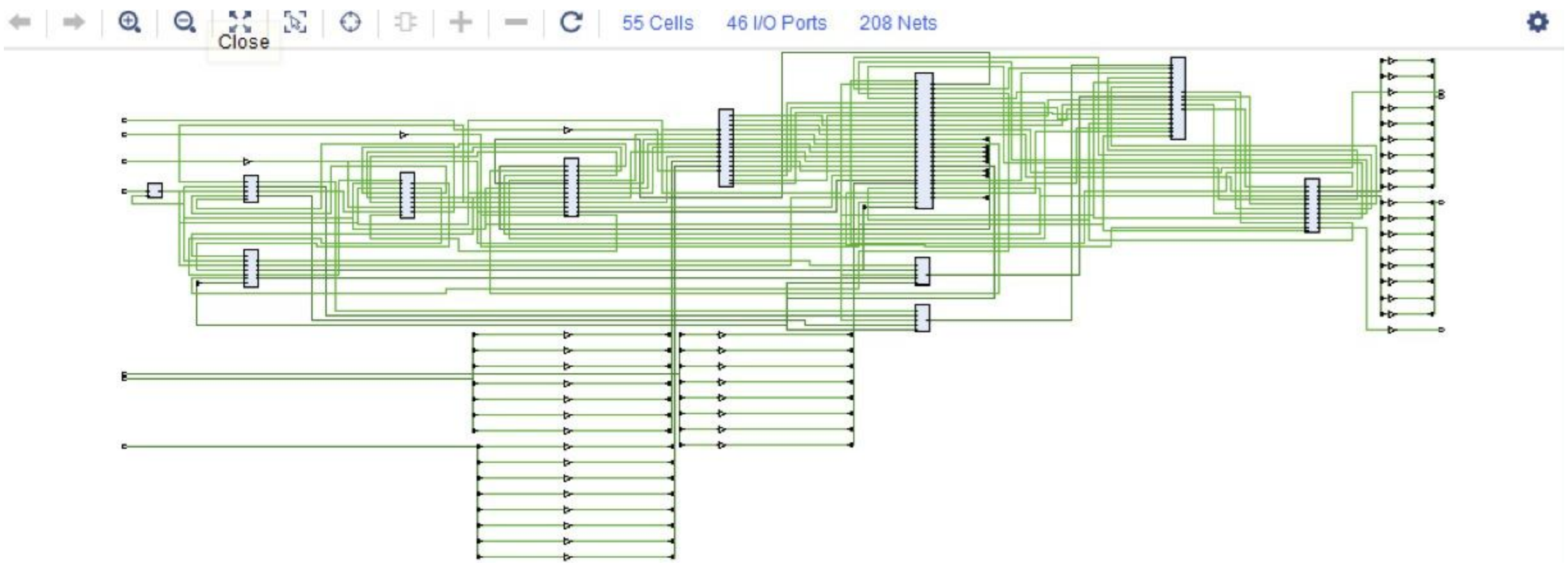
```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
MORESIMPLE REPORT:


############### BuiltIn -> RuleGroup=Design Read ###############
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
ID       Rule                    Alias                   Severity    File                                                                                  Line
================================================================================
[1]      DetectTopDesignUnits    DetectTopDesignUnits    Info        ../rtl/zigbee_phy_tx.v                                                                13
[0]      ElabSummary             ElabSummary             Info        ./spyglass-1/zigbee_phy_tx/lint/lint_rtl/spyglass_reports/SpyGlass/elab_summary.rpt  0
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

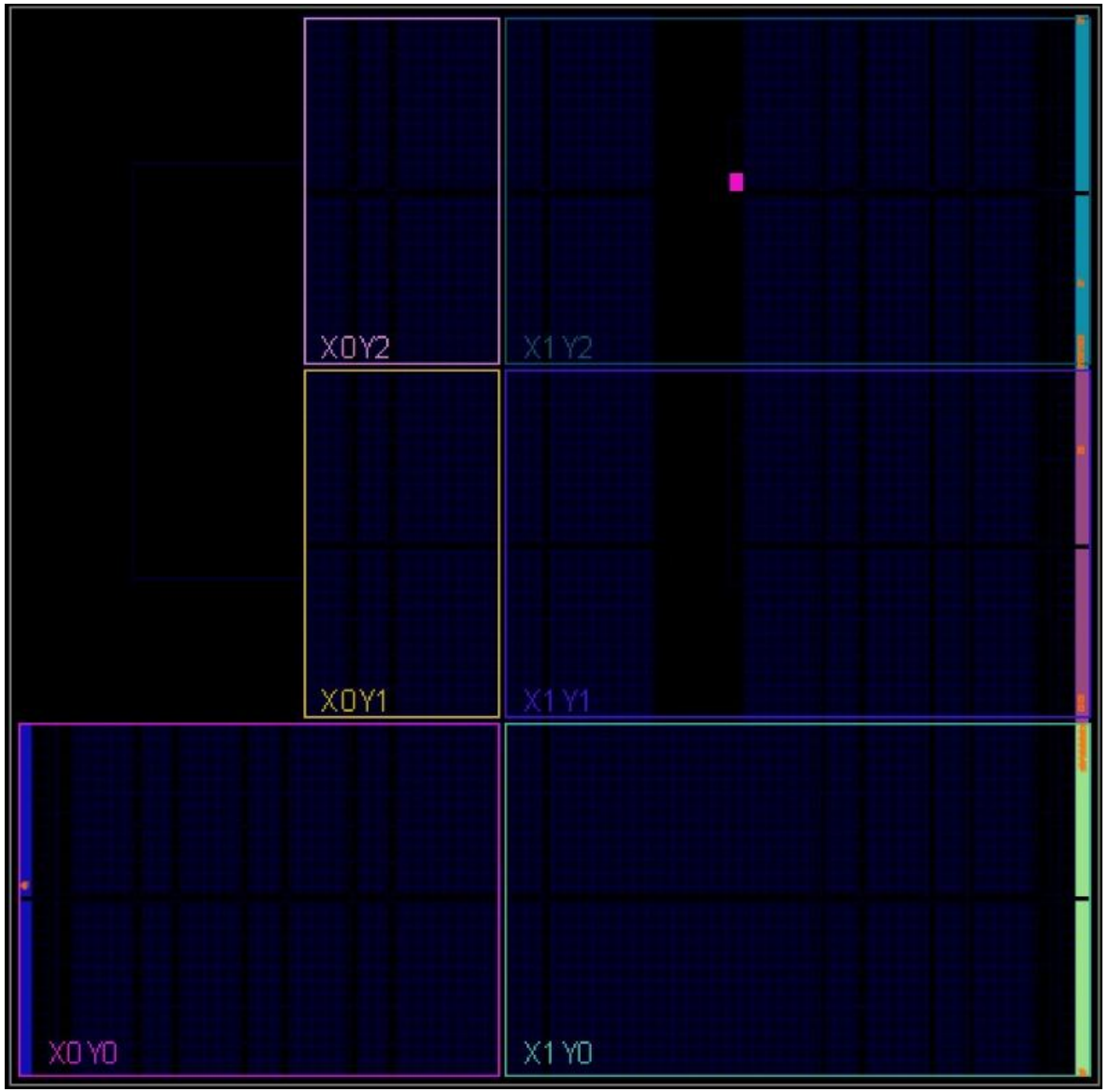

############### Non-BuiltIn -> Goal=lint/lint_rtl ###############
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
ID       Rule                Alias               Severity    File                        Line    Wt    Message
================================================================================
[5]      STARC05-2.11.3.1    SepStateNextLogic   Warning     ../rtl/chirp_modulation.v   582     10    Combinational and sequential parts of an FSM described in
[4]      STARC05-2.11.3.1    SepStateNextLogic   Warning     ../rtl/symbol_mapper.v      50      10    Combinational and sequential parts of an FSM described in
[3]      STARC05-2.11.3.1    SepStateNextLogic   Warning     ../rtl/zero_padder.v        46      10    Combinational and sequential parts of an FSM described in
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
```

FPGA

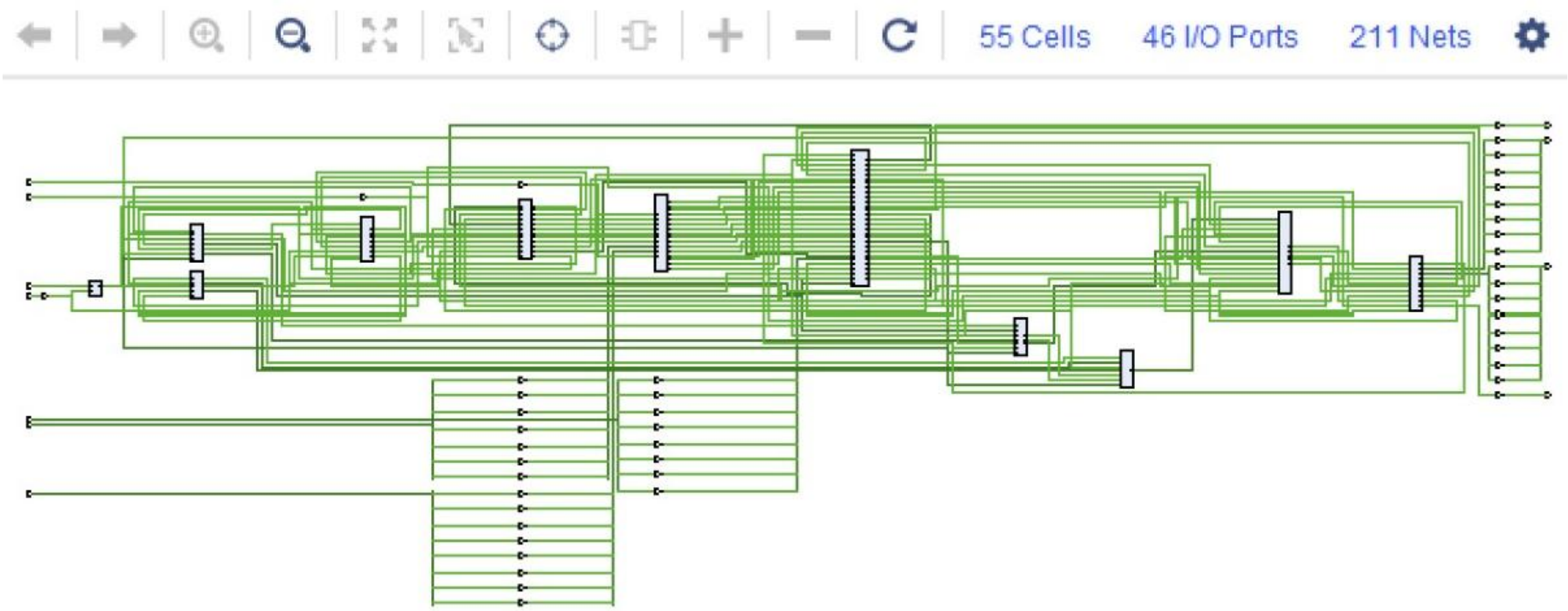
Synthesis



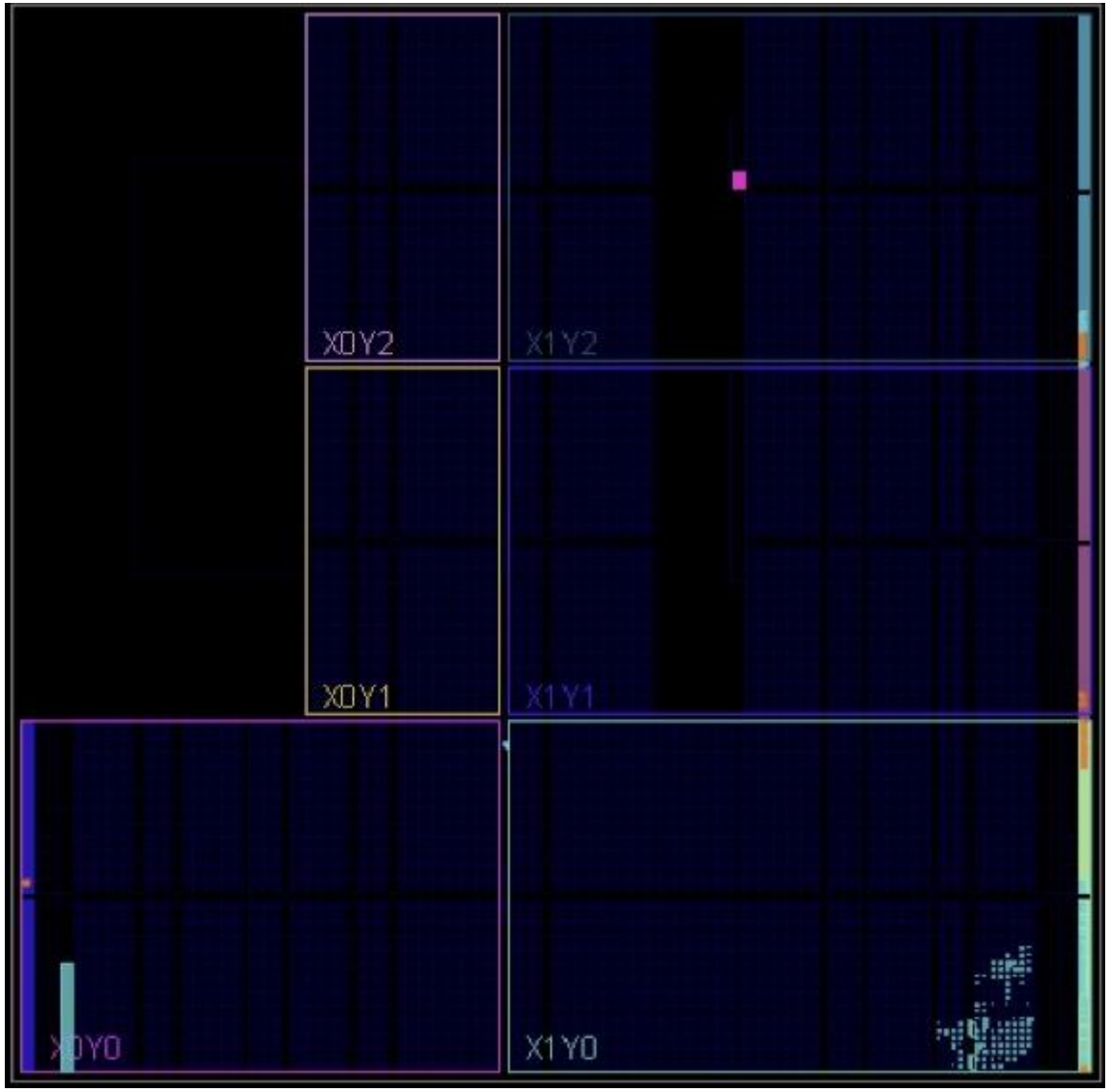
Synthesis



Implementation



Implementation



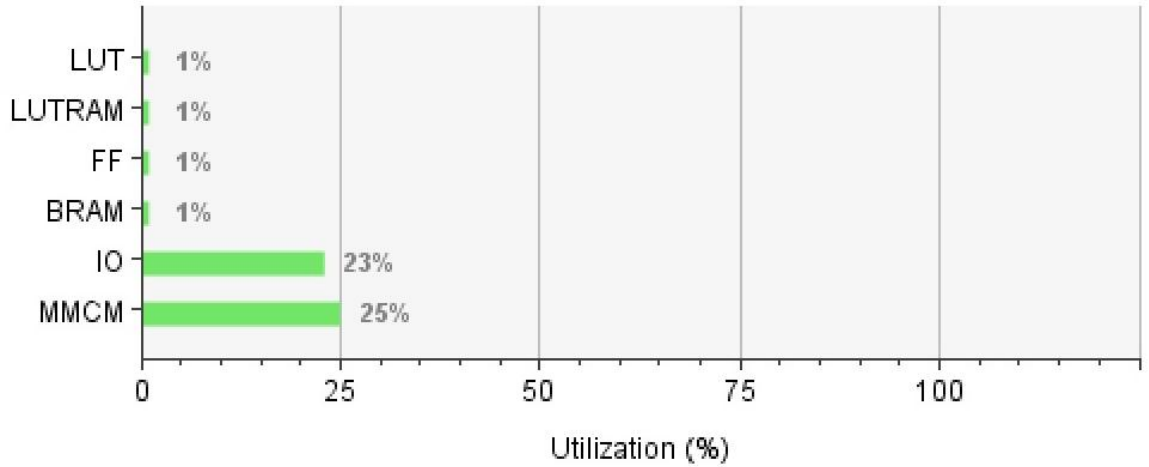
Reports

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	22.934 ns	Worst Hold Slack (WHS):	0.045 ns	Worst Pulse Width Slack (WPWS):	
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	
Total Number of Endpoints:	352	Total Number of Endpoints:	352	Total Number of Endpoints:	

All user specified timing constraints are met.

Name	Slice LUTs (53200)	Slice Registers (106400)	F7 Muxes (26600)	Slice (1330 0)	LUT as Logic (53200)	LUT as Memory (17400)	LUT Flip Flop Pairs (53200)	Block RAM Tile (140)	Bonded IOB (200)	BUFG (3
N zigbee_phy_tx	324	173	9	104	300	24	122	1	45	
u_clk_gen (clk_wiz_0)	0	0	0	0	0	0	0	0	0	
u_ctrl_fsm (master_ctr...	46	19	0	19	46	0	16	0	0	
u_demux (demux)	3	5	0	4	3	0	1	0	0	
u_form_ppdu (form_p...	11	15	0	8	11	0	10	0	0	
u_mod_top (qpsk_dqp...	150	63	9	48	150	0	43	0	0	
u_payload_ram (paylo...	26	0	0	7	2	24	0	0	0	
u_sm_I (symbol_map...	14	17	0	6	14	0	8	0	0	
u_sm_Q (symbol_ma...	10	17	0	6	10	0	8	0	0	
u_sm_ram_I (sm_ram)	1	3	0	1	1	0	0	0.5	0	
u_sm_ram_Q (sm_ra...	1	1	0	1	1	0	0	0.5	0	
u_zp (zero_padder)	62	33	0	21	62	0	22	0	0	

Reports



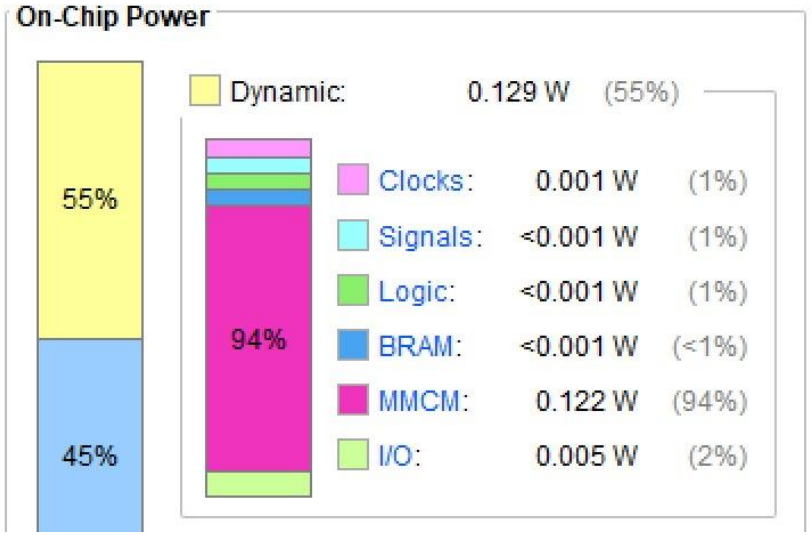
Summary

Resource	Utilization	Available	Utilization %
LUT	324	53200	0.61
LUTRAM	24	17400	0.14
FF	173	106400	0.16
BRAM	1	140	0.71
IO	45	200	22.50
MMCM	1	4	25.00

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	**0.237 W**
Design Power Budget:	**Not Specified**
Power Budget Margin:	**N/A**
Junction Temperature:	**27.7°C**
Thermal Margin:	57.3°C (4.8 W)
Effective ϑJA:	11.5°C/W
Power supplied to off-chip devices:	0 W



ASIC

Synthesis

Setup report

```
Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
zigbee_phy_tx      ForQA                 saed90nm_max
symbol_mapper_0    8000                  saed90nm_max
sm_ram_0           70000                 saed90nm_max

Point                                        Cap     Trans      Incr       Path
-------------------------------------------------------------------------------
clock clk (rise edge)                                          0.00       0.00
clock network delay (ideal)                                    0.00       0.00
u_sm_I/wr_addr_reg[0]/CLK (DFFX2)                      0.00    0.00 #     0.00 r
u_sm_I/wr_addr_reg[0]/QN (DFFX2)             2.86      0.17    0.45       0.45 f
u_sm_I/U7/ZN (INVX1)                         9.75      0.17    0.19       0.64 r
u_sm_I/wr_addr[0] (symbol_mapper_0)                            0.00       0.64 r
u_sm_ram_I/wr_addr[0] (sm_ram_0)                               0.00       0.64 r
u_sm_ram_I/U49/Q (AND2X1)                    5.42      0.13    1.43       2.06 r
u_sm_ram_I/U4108/Q (AND2X4)                  3.17      0.15    6.44       8.50 r
u_sm_ram_I/U319/Z (DELLN1X2)                 4.34      0.20    3.29      11.79 r
u_sm_ram_I/U58/ZN (INVX0)                    1.58      0.09    3.48      15.27 f
u_sm_ram_I/U59/ZN (INVX0)                    4.00      0.14    0.41      15.68 r
u_sm_ram_I/U3534/QN (NAND2X0)                4.58      0.23    3.29      18.97 f
u_sm_ram_I/U3304/ZN (INVX0)                  5.67      0.21    3.71      22.68 r
u_sm_ram_I/U318/Q (AO22X2)                   1.43      0.14    6.75      29.43 r
u_sm_ram_I/mem_reg[227][3]/D (DFFX1)                   0.14    0.29      29.72 r
data arrival time                                                        29.72

clock clk (rise edge)                                         31.25      31.25
clock network delay (ideal)                                    0.00      31.25
clock uncertainty                                             -1.25      30.00
u_sm_ram_I/mem_reg[227][3]/CLK (DFFX1)                         0.00      30.00 r
library setup time                                            -0.28      29.72
data required time                                                       29.72
-------------------------------------------------------------------------------
data required time                                                       29.72
data arrival time                                                       -29.72
-------------------------------------------------------------------------------
slack (MET)                                                               0.00
```

Synthesis

All violators report

```
1
2   ****************************************
3   Report : constraint
4            -all_violators
5   Design : zigbee_phy_tx
6   Version: O-2018.06-SP1
7   Date   : Thu Sep 17 02:34:47 2026
8   ****************************************
9
10  This design has no violated constraints.
11
12  1
```

Post synthesis formality

```
1  ****************************************************
2  Report          : failing_points
3
4  Reference       : ref:/WORK/zigbee_phy_tx
5  Implementation  : imp:/WORK/zigbee_phy_tx
6  Version         : L-2016.03-SP1
7  Date            : Thu Sep 17 17:22:32 2026
8  ****************************************************
9
10 No failing compare points.
11
12 1
```

```
1  ****************************************************
2  Report          : unverified_points
3
4  Reference       : ref:/WORK/zigbee_phy_tx
5  Implementation  : imp:/WORK/zigbee_phy_tx
6  Version         : L-2016.03-SP1
7  Date            : Thu Sep 17 17:22:32 2026
8  ****************************************************
9
10 No unverified compare points.
11
12 1
```

DFT

```
           Pattern Summary Report
-----------------------------------------------
#internal patterns                             0
-----------------------------------------------


    Uncollapsed Stuck Fault Summary Report
-----------------------------------------------
fault class                     code   #faults
------------------------------  ----  ---------
Detected                         DT     107856
Possibly detected                PT          0
Undetectable                     UD         11
ATPG untestable                  AU         17
Not detected                     ND          0
-----------------------------------------------
total faults                               107884
test coverage                               99.98%
--------------------------------|--------------
 Information: The test coverage above may be inferior
              than the real test coverage with customized
              protocol and test simulation library.
```

1

DFT

```
****************************************
Preview_dft report
For    : 'Insert_dft' command
Design : zigbee_phy_tx
Version: O-2018.06-SP1
Date   : Fri Sep 18 09:41:10 2026
****************************************

Number of chains: 4
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: no_mix

Chain      Scan Ports (si --> so)        # of Cells   Inst/Chain               Clock (port, time, edge)
-----      ----------------------        ----------   ----------               ------------------------
S 1        SI -->  SO                          811   u_ctrl_fsm/current_state_reg[0]
                          (scan_clk, 45.0, rising)
S 2        test_si2 -->  test_so4              811   u_payload_ram/mem_reg[89][2]
                          (scan_clk, 45.0, rising)
S 3        test_si3 -->  test_so2              811   u_sm_ram_I/mem_reg[115][3]
                          (scan_clk, 45.0, rising)
S 4        test_si4 -->  test_so3              811   u_sm_ram_Q/mem_reg[61][2]
                          (scan_clk, 45.0, rising)
1
```

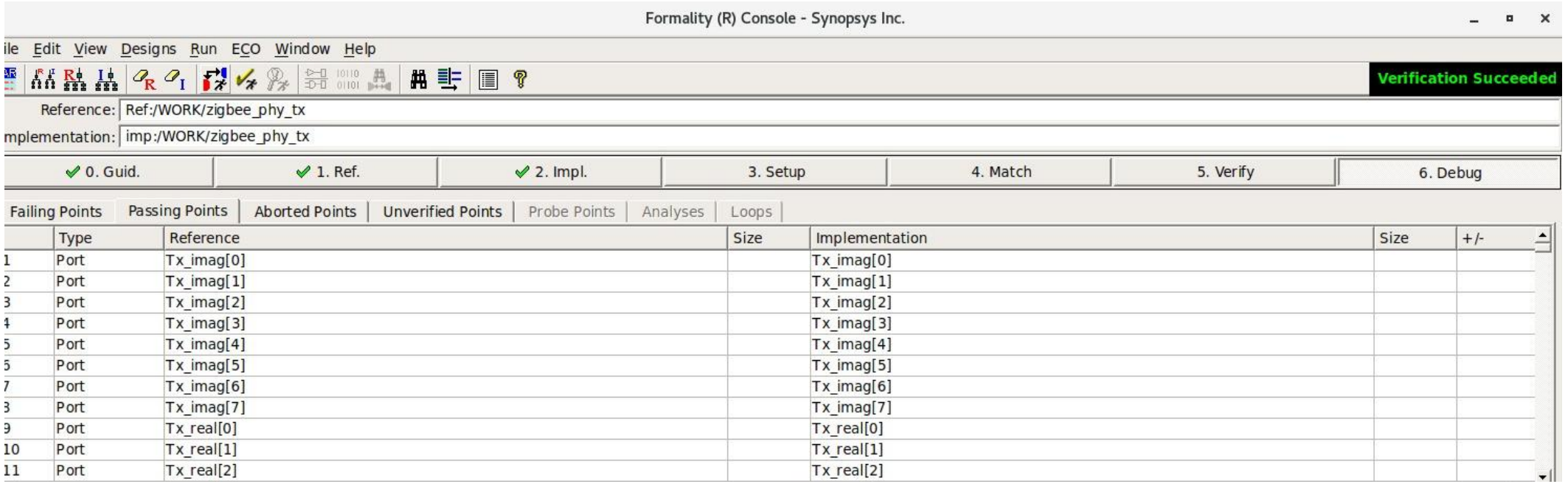
Post DFT formality

```
****************************************************
Report          : failing_points

Reference       : Ref:/WORK/zigbee_phy_tx
Implementation  : imp:/WORK/zigbee_phy_tx
Version         : L-2016.03-SP1
Date            : Fri Sep 18 06:14:05 2026
****************************************************

No failing compare points.

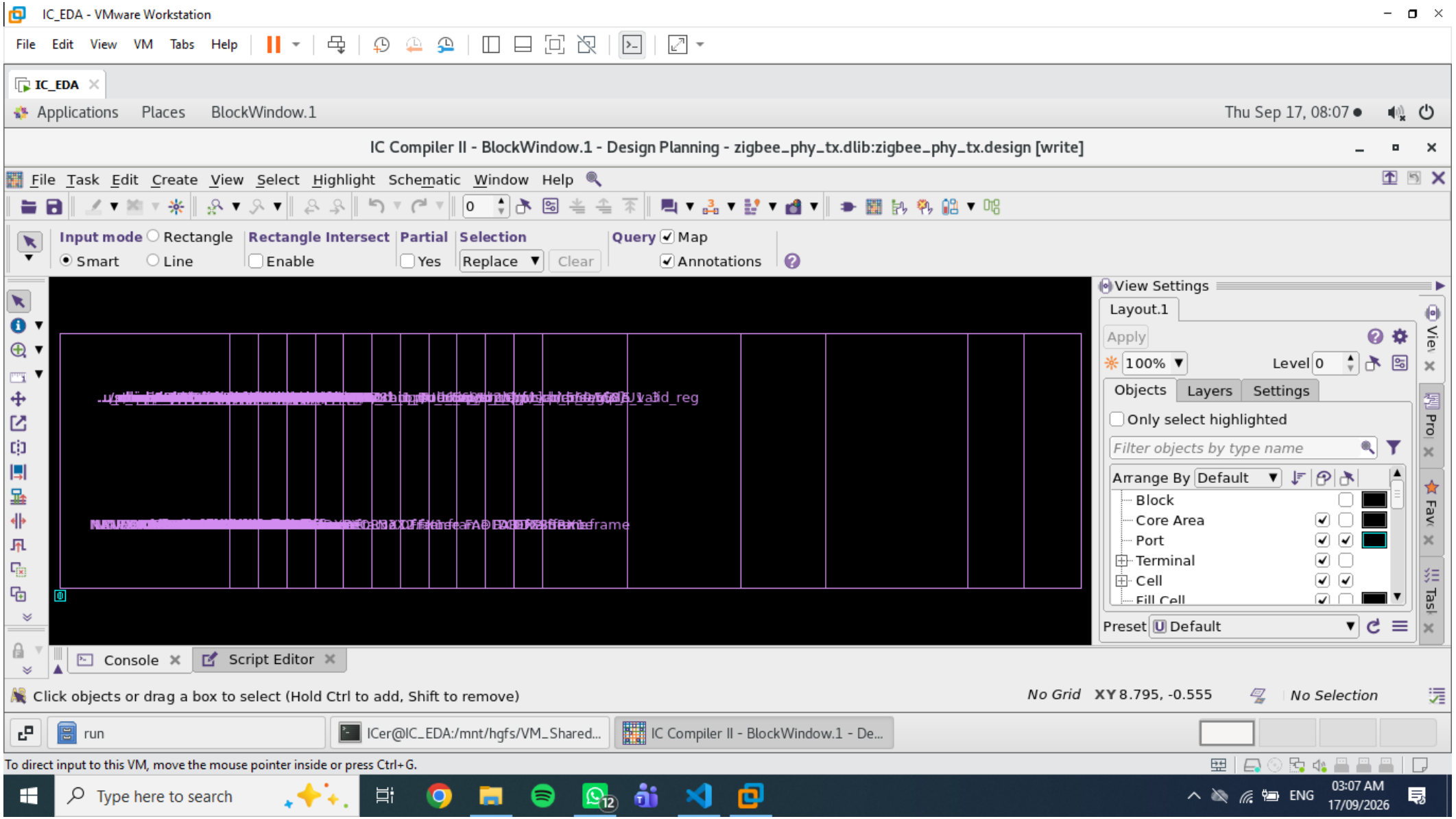
1
```



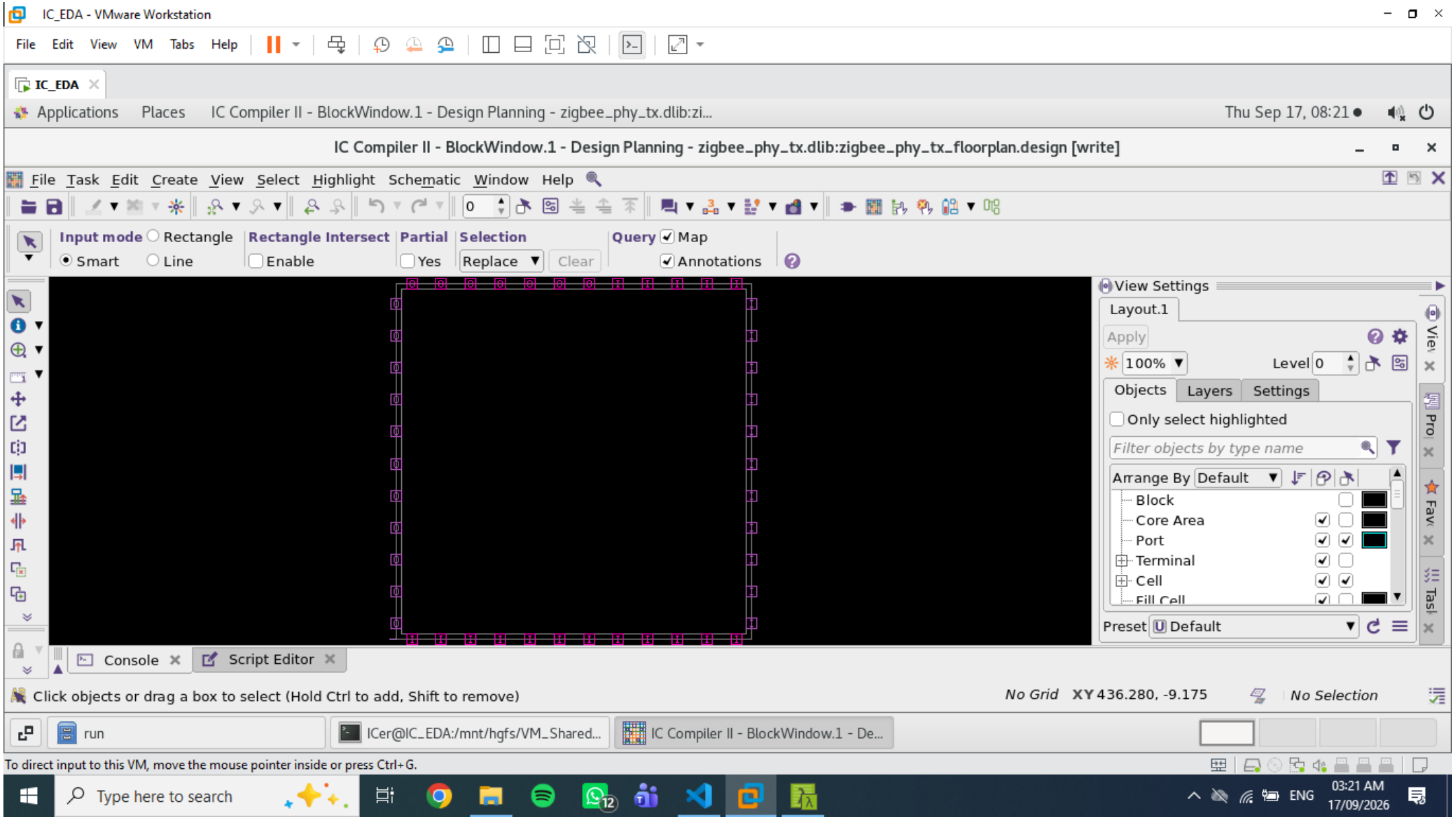
	Type	Reference	Size	Implementation	Size	+/-
1	Port	Tx_imag[0]		Tx_imag[0]		
2	Port	Tx_imag[1]		Tx_imag[1]		
3	Port	Tx_imag[2]		Tx_imag[2]		
4	Port	Tx_imag[3]		Tx_imag[3]		
5	Port	Tx_imag[4]		Tx_imag[4]		
6	Port	Tx_imag[5]		Tx_imag[5]		
7	Port	Tx_imag[6]		Tx_imag[6]		
8	Port	Tx_imag[7]		Tx_imag[7]		
9	Port	Tx_real[0]		Tx_real[0]		
10	Port	Tx_real[1]		Tx_real[1]		
11	Port	Tx_real[2]		Tx_real[2]		

PnR

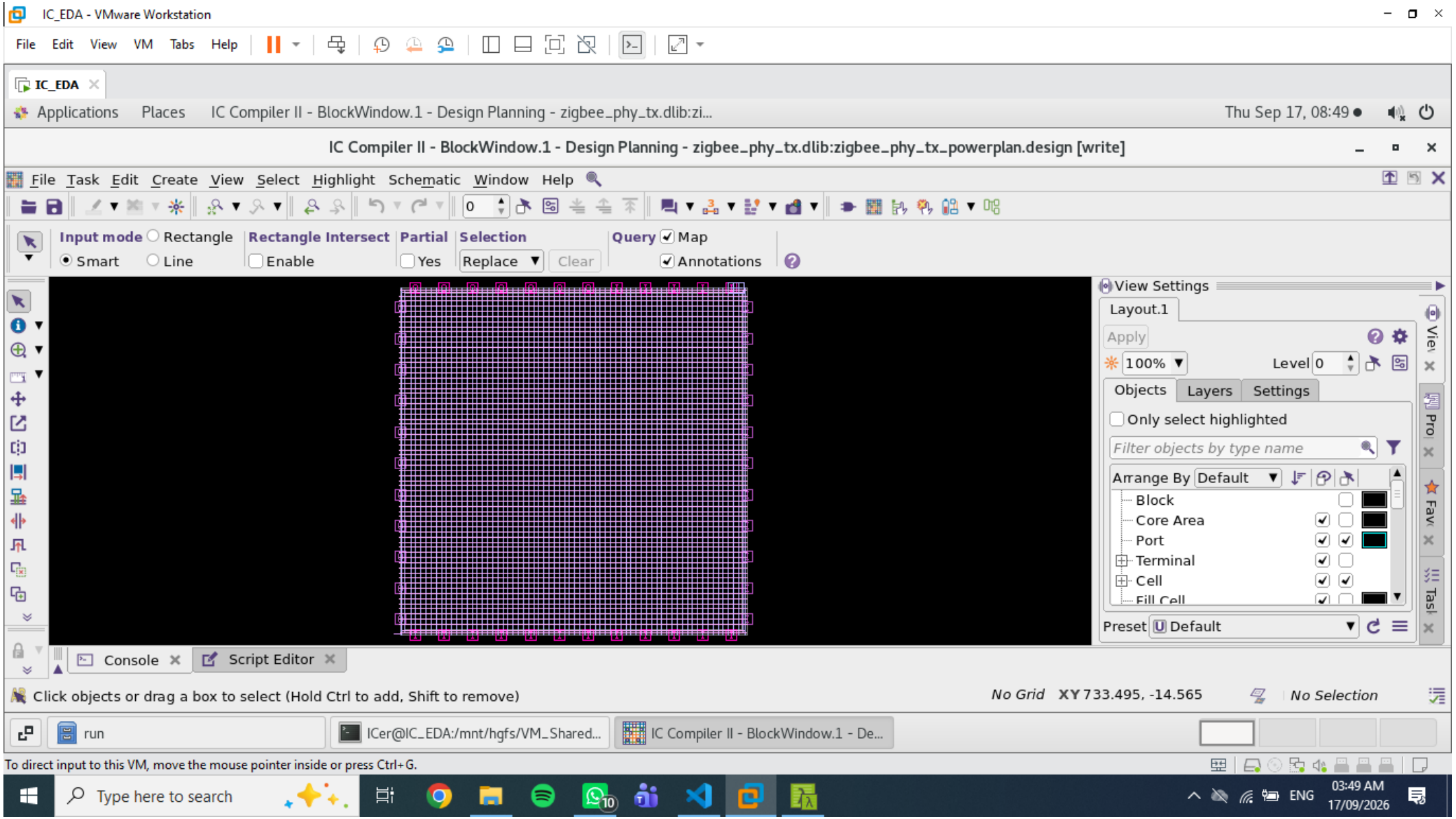
Design_lib


IC_EDA - VMware Workstation
File Edit View VM Tabs Help
IC_EDA
Applications Places BlockWindow.1
Thu Sep 17, 08:07
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Only select highlighted
Filter objects by type name
Arrange By Default
Block
Core Area
Port
Terminal
Cell
Fill Cell
Preset Default
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 8.795, -0.555
No Selection
run
ICer@IC_EDA:/mnt/hgfs/VM_Shared...
IC Compiler II - BlockWindow.1 - De...
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
03:07 AM
17/09/2026

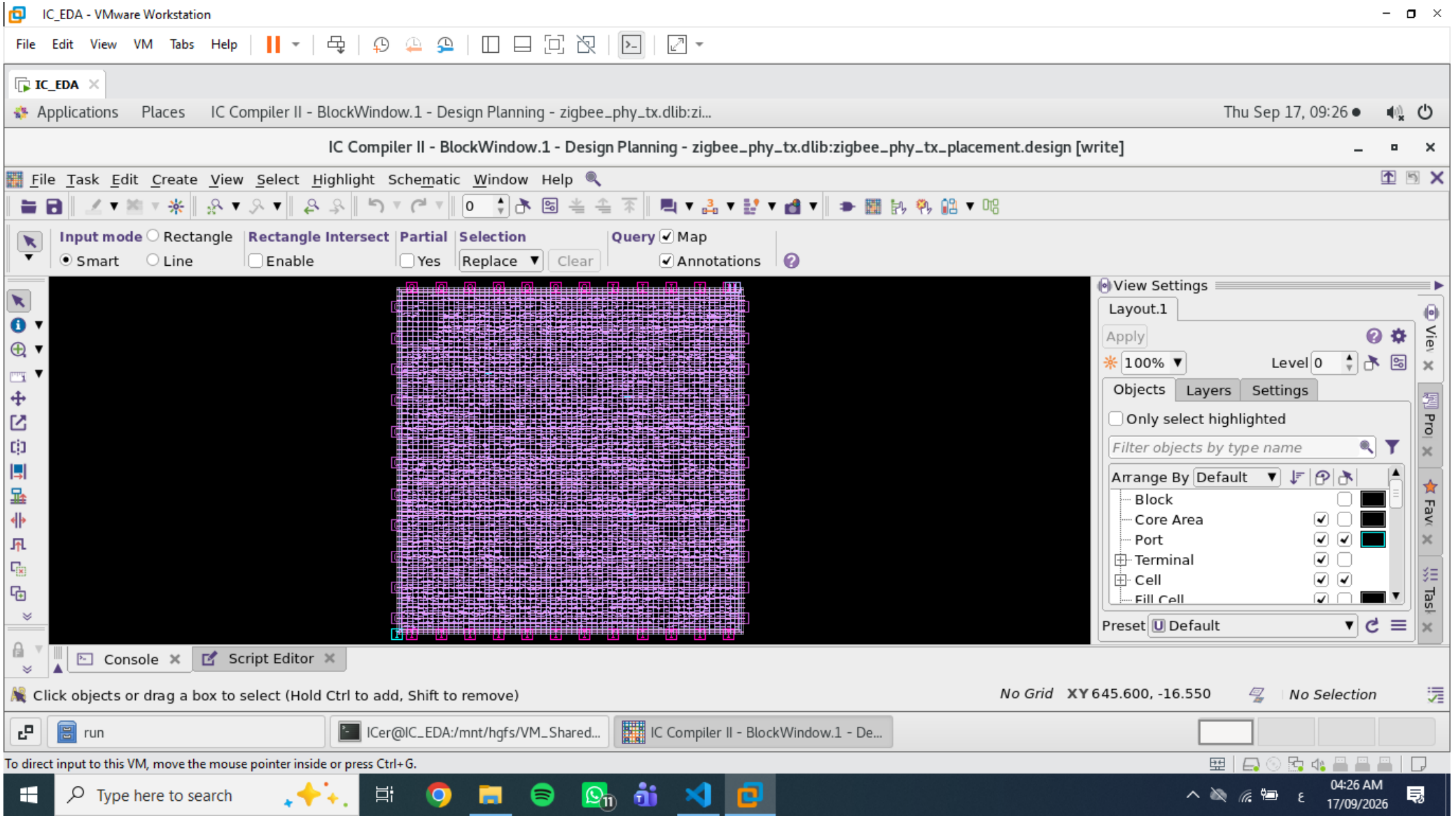
Floor_plan


IC_EDA - VMware Workstation
File Edit View VM Tabs Help
IC_EDA
Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zi...
Thu Sep 17, 08:21
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_floorplan.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Only select highlighted
Filter objects by type name
Arrange By Default
Block
Core Area
Port
Terminal
Cell
Fill Cell
Preset Default
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 436.280, -9.175
No Selection
run
ICer@IC_EDA:/mnt/hgfs/VM_Shared...
IC Compiler II - BlockWindow.1 - De...
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
03:21 AM
17/09/2026

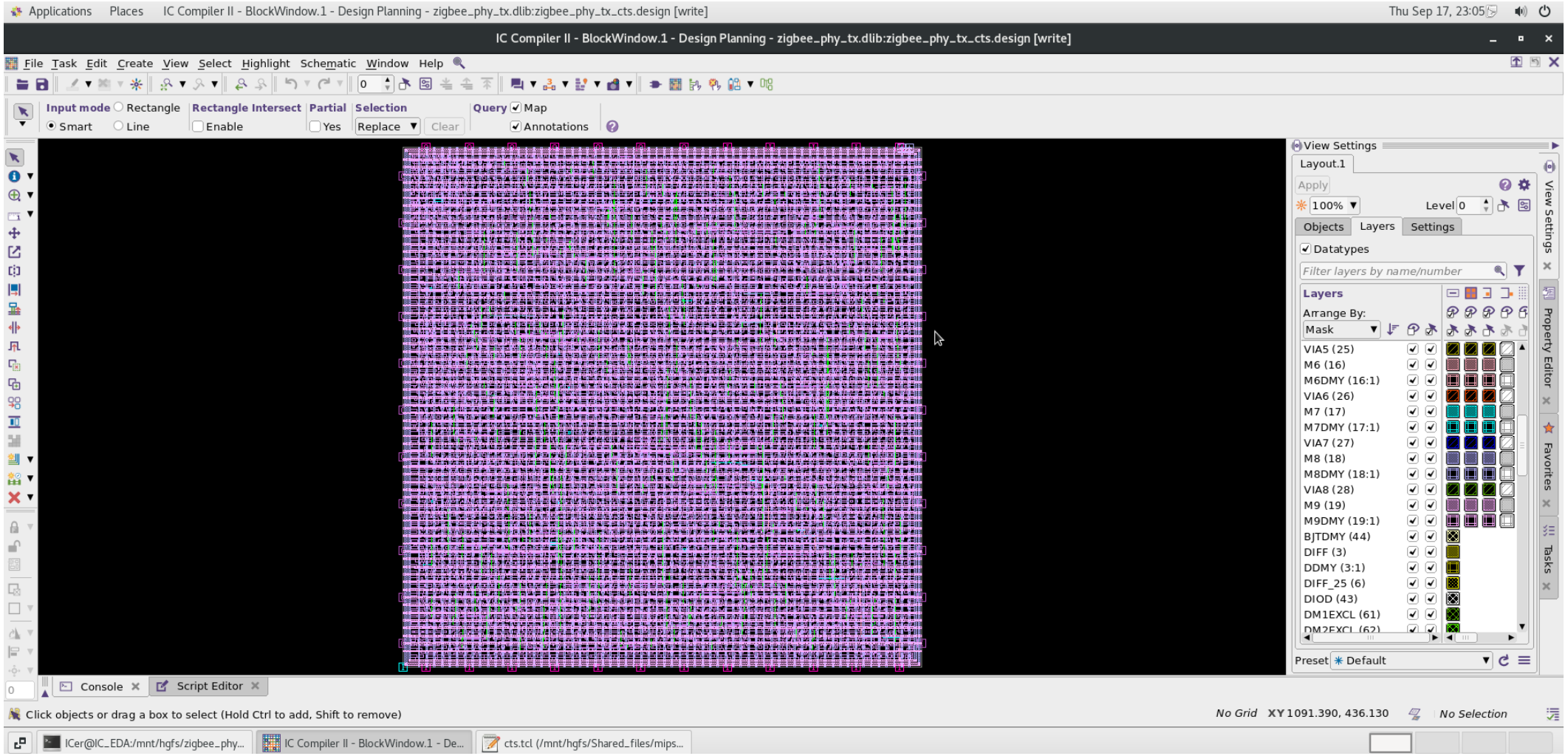
Power_plan


IC_EDA - VMware Workstation
File Edit View VM Tabs Help
IC_EDA
Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zi...
Thu Sep 17, 08:49
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_powerplan.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Only select highlighted
Filter objects by type name
Arrange By Default
Block
Core Area
Port
Terminal
Cell
Fill Cell
Preset Default
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 733.495, -14.565
No Selection
run
ICer@IC_EDA:/mnt/hgfs/VM_Shared...
IC Compiler II - BlockWindow.1 - De...
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
ENG
03:49 AM
17/09/2026

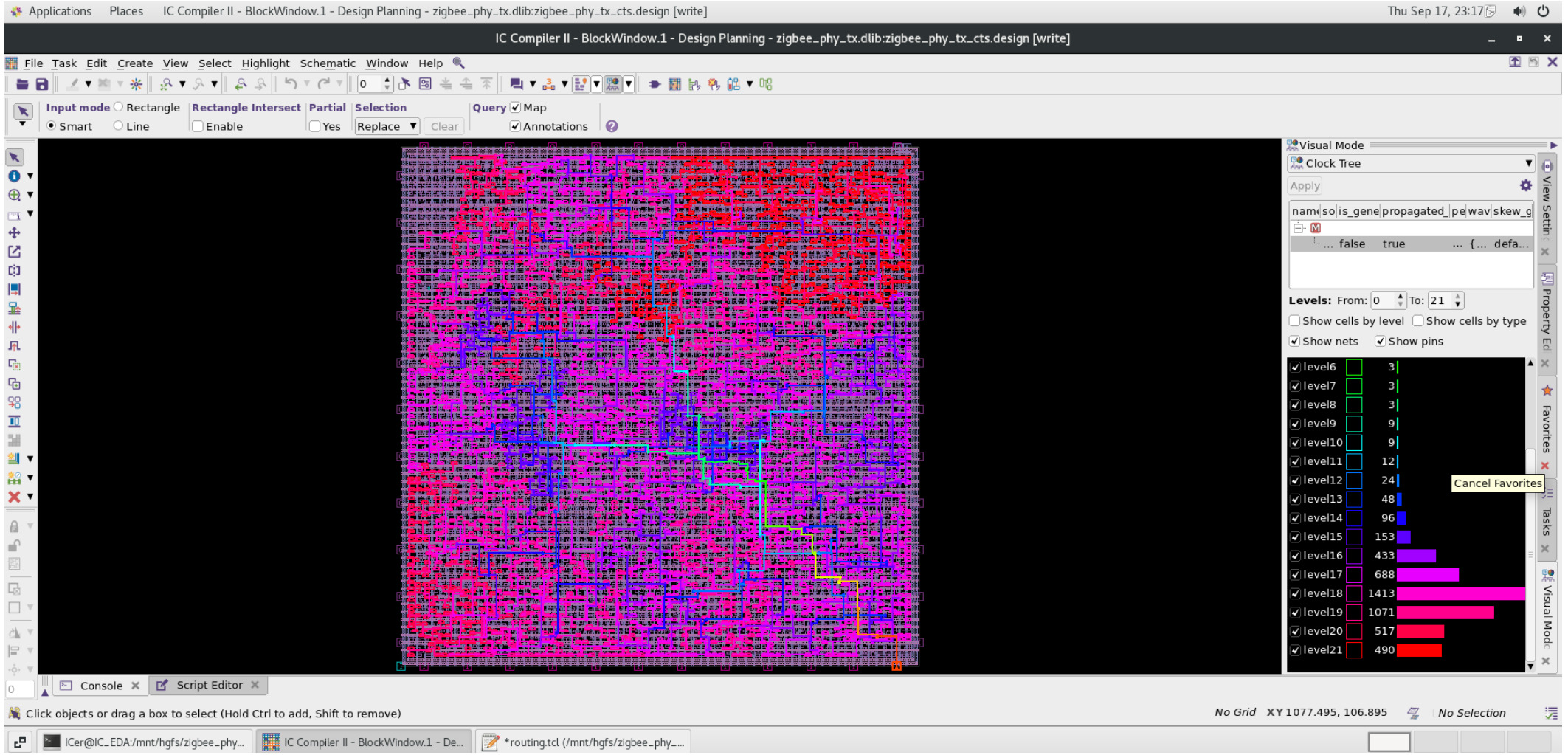
Placement


IC_EDA - VMware Workstation
File Edit View VM Tabs Help
IC_EDA
Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zi...
Thu Sep 17, 09:26
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_placement.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Only select highlighted
Filter objects by type name
Arrange By Default
Block
Core Area
Port
Terminal
Cell
Fill Cell
Preset Default
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 645.600, -16.550
No Selection
run
ICer@IC_EDA:/mnt/hgfs/VM_Shared...
IC Compiler II - BlockWindow.1 - De...
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
04:26 AM
17/09/2026

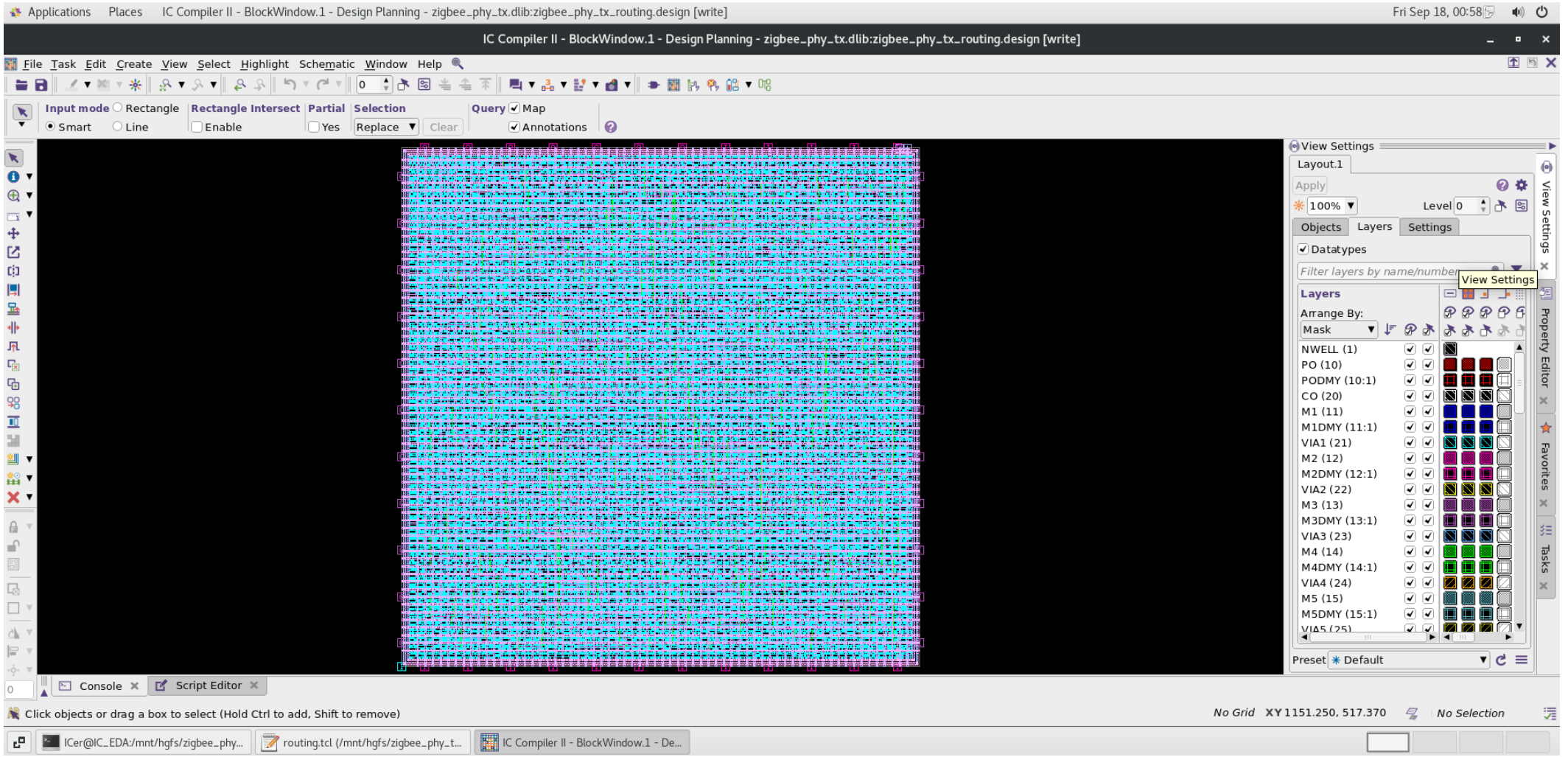
CTS


Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_cts.design [write]
Thu Sep 17, 23:05
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_cts.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Datatypes
Filter layers by name/number
Layers
Arrange By:
Mask
VIA5 (25)
M6 (16)
M6DMY (16:1)
VIA6 (26)
M7 (17)
M7DMY (17:1)
VIA7 (27)
M8 (18)
M8DMY (18:1)
VIA8 (28)
M9 (19)
M9DMY (19:1)
BJTDMY (44)
DIFF (3)
DDMY (3:1)
DIFF_25 (6)
DIOD (43)
DM1EXCL (61)
DM2EXCL (62)
Preset Default
View Settings
Property Editor
Favorites
Tasks
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 1091.390, 436.130
No Selection
ICer@IC_EDA:/mnt/hgfs/zigbee_phy...
IC Compiler II - BlockWindow.1 - De...
cts.tcl (/mnt/hgfs/Shared_files/mips...

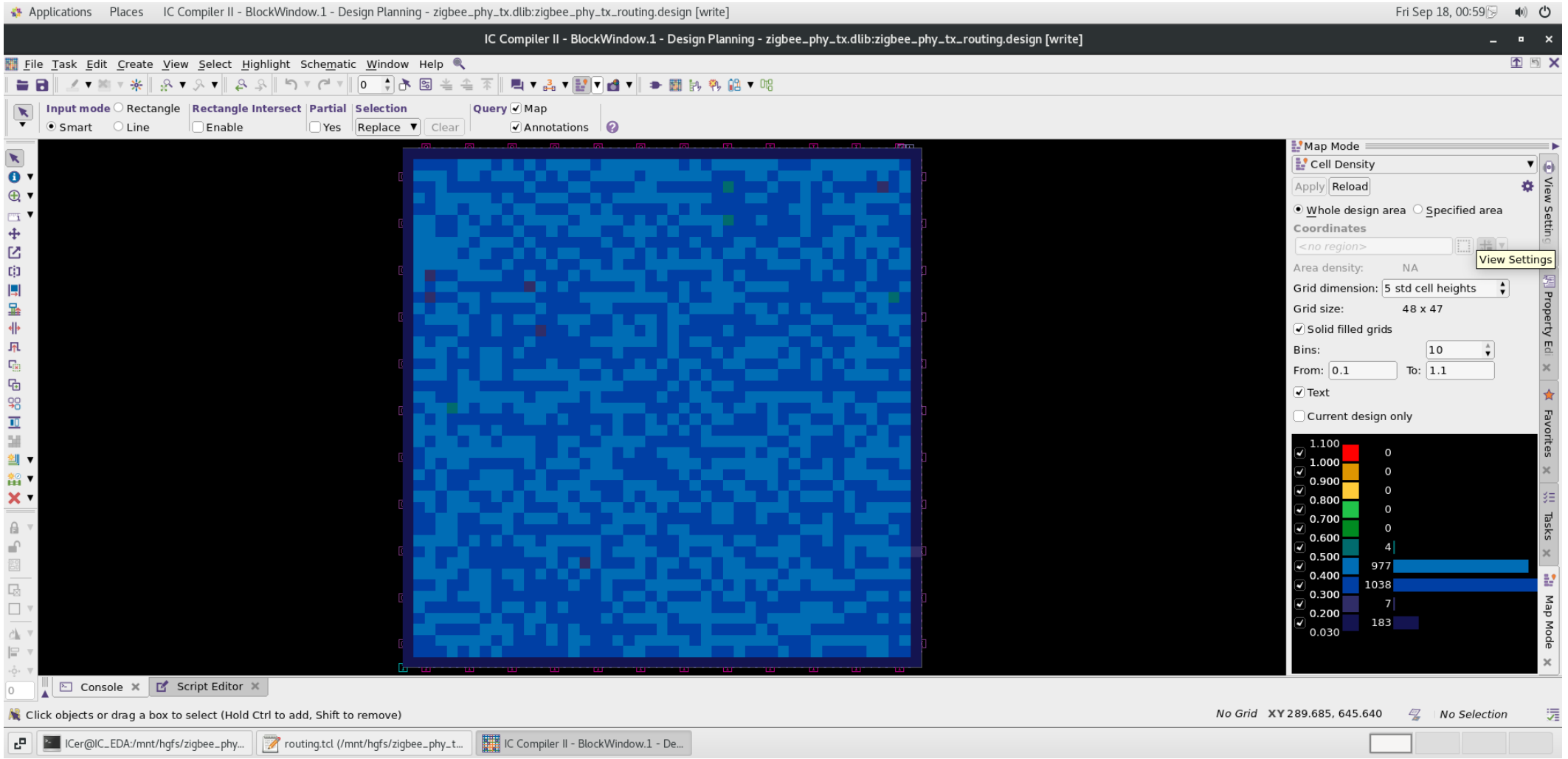
CTS


Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_cts.design [write]
Thu Sep 17, 23:17
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_cts.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
Visual Mode
Clock Tree
Apply
Levels: From: 0 To: 21
Show cells by level
Show cells by type
Show nets
Show pins
level6 3
level7 3
level8 3
level9 9
level10 9
level11 12
level12 24
level13 48
level14 96
level15 153
level16 433
level17 688
level18 1413
level19 1071
level20 517
level21 490
Cancel Favorites
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 1077.495, 106.895
No Selection

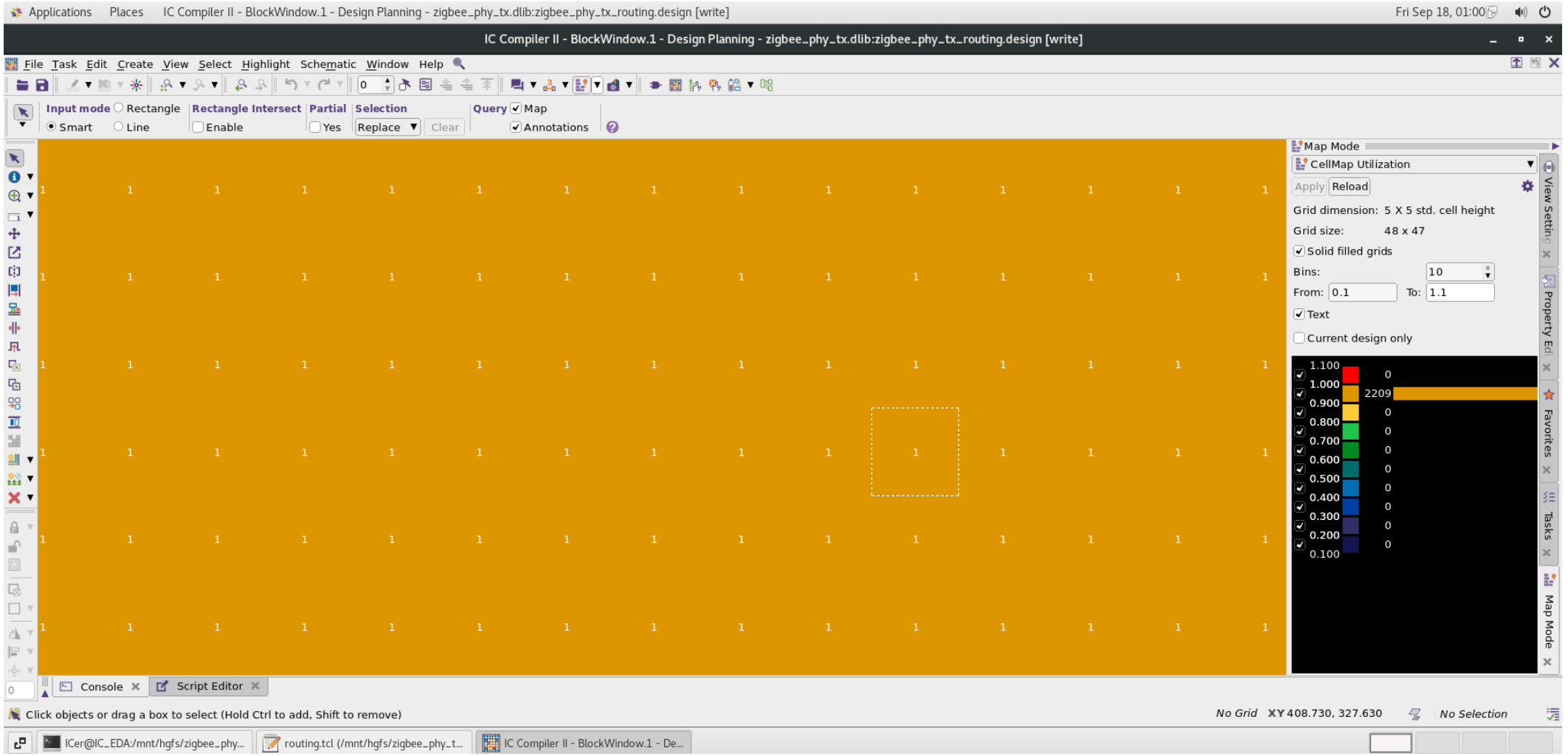
Routing


Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
Fri Sep 18, 00:58
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Smart Rectangle Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
View Settings
Layout.1
Apply
100%
Level 0
Objects Layers Settings
Datatypes
Filter layers by name/number
View Settings
Layers
Arrange By:
Mask
NWELL (1)
PO (10)
PODMY (10:1)
CO (20)
M1 (11)
M1DMY (11:1)
VIA1 (21)
M2 (12)
M2DMY (12:1)
VIA2 (22)
M3 (13)
M3DMY (13:1)
VIA3 (23)
M4 (14)
M4DMY (14:1)
VIA4 (24)
M5 (15)
M5DMY (15:1)
Preset Default
View Settings
Property Editor
Favorites
Tasks
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 1151.250, 517.370
No Selection
ICer@IC_EDA:/mnt/hgfs/zigbee_phy...
routing.tcl (/mnt/hgfs/zigbee_phy_t...
IC Compiler II - BlockWindow.1 - De...

Routing


Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
Fri Sep 18, 00:59
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Smart Rectangle Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
Map Mode
Cell Density
Apply Reload
Whole design area Specified area
Coordinates
<no region>
View Settings
Area density: NA
Grid dimension: 5 std cell heights
Grid size: 48 x 47
Solid filled grids
Bins: 10
From: 0.1 To: 1.1
Text
Current design only
1.100 0
1.000 0
0.900 0
0.800 0
0.700 0
0.600 0
0.500 4
0.400 977
0.300 1038
0.200 7
0.030 183
View Setting
Property Ed
Favorites
Tasks
Map Mode
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 289.685, 645.640
No Selection
ICer@IC_EDA:/mnt/hgfs/zigbee_phy...
routing.tcl (/mnt/hgfs/zigbee_phy_t...
IC Compiler II - BlockWindow.1 - De...

Routing


Applications Places IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
Fri Sep 18, 01:00
IC Compiler II - BlockWindow.1 - Design Planning - zigbee_phy_tx.dlib:zigbee_phy_tx_routing.design [write]
File Task Edit Create View Select Highlight Schematic Window Help
Input mode Rectangle Smart Line
Rectangle Intersect Enable
Partial Yes
Selection Replace Clear
Query Map Annotations
Map Mode
CellMap Utilization
Apply Reload
Grid dimension: 5 X 5 std. cell height
Grid size: 48 x 47
Solid filled grids
Bins: 10
From: 0.1 To: 1.1
Text
Current design only
1.100 1.000 0.900 0.800 0.700 0.600 0.500 0.400 0.300 0.200 0.100
0 2209 0 0 0 0 0 0 0 0
View Setting
Property Ed
Favorites
Tasks
Map Mode
Console
Script Editor
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)
No Grid XY 408.730, 327.630
No Selection
ICer@IC_EDA:/mnt/hgfs/zigbee_phy...
routing.tcl (/mnt/hgfs/zigbee_phy_t...
IC Compiler II - BlockWindow.1 - De...

Reports

zigbee_phy_tx / pnr / 7_routing / results / reports / **routing_violations.rpt**

KhaledGamal1 GDS generated

Code | Blame | 30 lines (21 loc) · 886 Bytes

```
1   ****************************************
2   Report : constraint
3            -all_violators
4   Design : zigbee_phy_tx
5   Version: O-2018.06-SP1
6   Date   : Fri Sep 18 02:39:34 2026
7   ****************************************
8
9      late_timing
10     -----------
11
12  Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)
13  Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
14  ---------------------------------------------------------------------------------------------------
15  No paths.
16
17     early_timing
18     -----------
19
20  Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)
21  Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
22  ---------------------------------------------------------------------------------------------------
23  No paths.
24
25
26
27
28
29     Total number of violation(s): 0
30  1
```

Reports

zigbee_phy_tx / pnr / 7_routing / results / reports / **qor.rpt**

KhaledGamal1 GDS generated

Code | Blame | 24 lines (21 loc) · 960 Bytes

```
1   ****************************************
2   Report : qor
3           -summary
4   Design : zigbee_phy_tx
5   Version: O-2018.06-SP1
6   Date   : Fri Sep 18 02:39:35 2026
7   ****************************************
8   Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)
9
10  Timing
11  --------------------------------------------------------------------------
12  Context                              WNS            TNS            NVE
13  --------------------------------------------------------------------------
14  Design             (Setup)          18.22           0.00              0
15
16  Design             (Hold)            0.00           0.00              0
17  --------------------------------------------------------------------------
18
19  Miscellaneous
20  --------------------------------------------------------------------------
21  Cell Area (netlist):                         172107.88
22  Cell Area (netlist and physical only):       431386.21
23  Nets with DRC Violations:        0
24  1
```

Reports

zigbee_phy_tx / pnr / 7_routing / results / reports / **pg_connectivity.rpt**

KhaledGamal1 Routing and GUI view

Code | Blame | 31 lines (31 loc) · 1.04 KB

```
Loading cell instances...
Number of Standard Cells: 96136
Number of Macro Cells: 0
Number of IO Pad Cells: 0
Number of Blocks: 0
Loading P/G wires and vias...
Number of VDD Wires: 185
Number of VDD Vias: 27790
Number of VDD Terminals: 132
Number of VSS Wires: 182
Number of VSS Vias: 26692
Number of VSS Terminals: 128
**************Verify net VDD connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
**************Verify net VSS connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Overall runtime: 1 seconds.
```

END OF PRESENTATION

Thank You!

Questions & Open Discussion